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Aerial Image Formation – The Basics

Projection printing means projecting the image of a photomask (also called a reticle) onto a resist-coated wafer. Projection imaging tools are sophisticated reduction cameras with stages that allow, through a combination of stepping or stepping and scanning motions, exposure of many (reduced) copies of a mask pattern onto a large wafer. The image of the mask that is projected into the photoresist defines the information content used by the photoresist to form the final resist image. Understanding the limits and capabilities of projection imaging is the first step in understanding the limits and capabilities of lithography.

The imaging process is a well-studied optical phenomenon. We will begin at the most basic level with a mathematical description of light, followed by the theory of image formation. The impact of partial coherence will then be added. In the following chapter, nonideal and vector descriptions of image formation will be added to the basic description provided here.

2.1 Mathematical Description of Light

A mathematical description of light must, of course, begin with Maxwell's equations. However, these fundamental equations can be quite cumbersome and general solutions under diverse boundary conditions almost always entail detailed numerical calculations. However, under some circumstances a simplified form of Maxwell's equations, in particular what is called the wave equation, can be used and often allows reasonably compact solutions. In the case of imaging theory, decomposing an arbitrary wave into a summation of plane waves provides the simplest approach for calculating the images projected into a photoresist-coated wafer.

2.1.1 Maxwell's Equations and the Wave Equation

Light is an electromagnetic wave with coupled electric and magnetic fields traveling through space and the materials that occupy that space. Maxwell's equations describe two fundamental, coupled fields, the electric field $\mathbf{E}$ (units of volts per meter) and the magnetic field $\mathbf{H}$ (units of amperes per meter), both vector quantities. Further, the effect of these fields on a material can give rise to four other quantities: three vectors, the electric displacement $\mathbf{D}$ (units of coulomb per square meter), the magnetic induction $\mathbf{B}$ (units of tesla, or webers per square meter) and the electric current density $\mathbf{J}$ (units of amps per square meter), and the scalar electric charge density ρ (units of coulomb per cubic meter). These six quantities are related by the four Maxwell's equations (shown here in MKS or SI units):

$$\begin{aligned}
 \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t} &= \mathbf{J} && \text{(Maxwell's extension of Ampere's law)} \\
 \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= 0 && \text{(Faraday's law of induction)} \\
 \nabla \cdot \mathbf{D} &= \rho && \text{(Gauss's law)} \\
 \nabla \cdot \mathbf{B} &= 0
 \end{aligned} \tag{2.1}$$

where the standard curl and divergence operators are used (see Appendix B). The first two equations above show how a time-varying electric field (or displacement) causes a change in the magnetic field (or induction), and vice versa. The second two equations show how static charge affects the electric and magnetic fields (the last equation set to zero is a statement that there are no magnetic monopoles and therefore no net magnetic charge). A combination of Ampere's law and Gauss's law leads to the continuity equation, an expression of the conservation of electrical charge.

Additionally, the properties of the materials involved provide relationships between the current density and the electric field, the electric displacement and the electric field, and the magnetic induction and the magnetic field. In general, these relationships can be quite complicated, but in many circumstances they become simple. If the material that the electromagnetic radiation is propagating through is moving slowly relative to the speed of light, and the fields involved are time-harmonic (a concept that will be defined below), then the three material equations become

$$\begin{aligned}
 \mathbf{J} &= \sigma \mathbf{E} \\
 \mathbf{D} &= \epsilon \mathbf{E} \\
 \mathbf{B} &= \mu \mathbf{H}
 \end{aligned} \tag{2.2}$$

where σ is the conductivity of the material, ϵ is its dielectric constant (sometimes called the electrical permittivity) and μ is its magnetic permeability, all of which are frequency dependent. Over the range of electric and magnetic field strengths of interest here, these three material properties can be considered constants. The first of the three material equations in Equation (2.2) is the differential form of Ohm's law. Note that for perfectly transparent materials (also called dielectrics), $\sigma = 0$. Thus, absorption in a material is

related to the generation of current density under the influence of an electric field (i.e. conduction). Also, in general, the materials of interest to the lithographer are nonmagnetic, and thus $\mu = \mu_0$, the magnetic permeability of free space.

Consider a further simplification where the materials involved are homogeneous (the three material properties are not a function of position). In that case, the three material properties σ , ϵ and μ , become simple numbers (scalars), rather than tensors. By using these material relationships, the first two Maxwell's equations can be rewritten in terms of electric and magnetic fields only.

$$\begin{aligned}\nabla \times \mathbf{H} - \epsilon \frac{\partial \mathbf{E}}{\partial t} &= \sigma \mathbf{E} \\ \nabla \times \mathbf{E} + \mu \frac{\partial \mathbf{H}}{\partial t} &= 0\end{aligned}\tag{2.3}$$

The goal now will be to take these two coupled first-order differential equations and create two noncoupled second-order equations. When these materials are free of charge ($\rho = 0$), $\nabla \cdot \mathbf{E} = 0$ so that

$$\nabla^2 \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla \times (\nabla \times \mathbf{E}) = -\nabla \times (\nabla \times \mathbf{E})\tag{2.4}$$

Taking the curl of one of Equation (2.3) and substituting in the other,

$$\begin{aligned}\nabla^2 \mathbf{E} &= \epsilon \mu \frac{\partial^2 \mathbf{E}}{\partial t^2} + \mu \sigma \frac{\partial \mathbf{E}}{\partial t} \\ \nabla^2 \mathbf{H} &= \epsilon \mu \frac{\partial^2 \mathbf{H}}{\partial t^2} + \mu \sigma \frac{\partial \mathbf{H}}{\partial t}\end{aligned}\tag{2.5}$$

As a final simplification, for a nonabsorbing medium ($\sigma = 0$), these equations become classic wave equations:

$$\begin{aligned}\nabla^2 \mathbf{E} &= \frac{1}{v^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} \\ \nabla^2 \mathbf{H} &= \frac{1}{v^2} \frac{\partial^2 \mathbf{H}}{\partial t^2}\end{aligned}\tag{2.6}$$

where $v = 1/\sqrt{\epsilon\mu}$, the velocity of the wave. Letting ϵ_0 and μ_0 be the values of these material properties in vacuum, the speed of light in vacuum is then $c = 1/\sqrt{\epsilon_0\mu_0}$. These constants have values of

$$\begin{aligned}c &= 2.99792458 \times 10^8 \text{ m/s} \\ \epsilon_0 &= 8.8541878 \times 10^{-12} \text{ farads/m (1 farad = 1 coulomb/volt)} = 1/(4\pi c^2) \times 10^7 \text{ farads/m} \\ \mu_0 &= 4\pi \times 10^{-7} \text{ henries/m (1 henry = 1 weber/amp = 1 volt-second/amp)}\end{aligned}$$

From this, the refractive index of a material can be defined as the ratio of the speed of light in vacuum to that in the material:

$$n \equiv \frac{c}{v} = \sqrt{\frac{\epsilon\mu}{\epsilon_0\mu_0}}\tag{2.7}$$

It is sometimes useful to define the dielectric constant and magnetic permeability relative to the free-space values:

$$\epsilon_r \equiv \frac{\epsilon}{\epsilon_0}, \quad \mu_r \equiv \frac{\mu}{\mu_0} \quad (2.8)$$

so that

$$n = \sqrt{\epsilon_r \mu_r} \quad (2.9)$$

And since we are mostly concerned with nonmagnetic materials with $\mu_r = 1$, the refractive index will generally be just the square root of the relative dielectric constant of the material (remembering that we have so far only looked at materials where $\sigma = 0$).

2.1.2 General Harmonic Fields and the Plane Wave in a Nonabsorbing Medium

Since the electric and magnetic fields are related to each other (the magnetic field is always perpendicular to the electric field and both fields are always perpendicular to the direction of propagation, as will be proven in section 2.1.3) and since photoresist reacts chemically to the magnitude of the electric field only, we can usually describe light by considering just the electric field. A general harmonic electric field vector $\mathbf{E}$ (due to monochromatic light of frequency ω) at any point P and time t can be described by a deceptively simple-looking sinusoidal equation:

$$\mathbf{E}(P, t) = \mathbf{A}(P) \cos(\omega t + \Phi(P)) \quad (2.10)$$

where $\mathbf{A}$ is the amplitude (like the electric field, a vector) and Φ is the phase, both of which are position dependent, in general, and will depend on initial and boundary conditions. As an example of one form of Equation (2.10), consider a ‘plane wave’ of light traveling in the $+z$ direction. The term plane wave refers to the shape of the *wavefront*, i.e. the shape of the function $\Phi(P) = \text{constant}$. Thus, a plane wave traveling in the $+z$ direction would require a constant phase (and, consequently, a constant amplitude) in the x - y plane. Such a plane wave would be described by the equation

$$E(P, t) = A \cos(\omega t - kz + \delta) \quad (2.11)$$

where k is a constant called the *propagation constant* or the *wave number* and δ is a phase offset.

How does the wave represented by Equation (2.11) propagate? We can think of the wave as having a certain shape, and this shape travels through time. Thus, this wave will have the given shape at all points in space and time such that $\omega t - kz = \text{constant}$, giving the same electric field. In other words, the wavefront (a plane in this case) travels through space and time according to

$$z - z_0 = \frac{\omega}{k} t \quad (2.12)$$

where z_0 is a constant corresponding to the position of the plane wave at $t = 0$. This is simply a plane of light traveling in the $+z$ direction at speed ω/k . Thus, since

$$v = \frac{c}{n} = \frac{\omega}{k} \quad (2.13)$$

and since $c = \omega\lambda/2\pi$ (a general property of waves), we obtain

$$k = \omega\sqrt{\epsilon\mu} = 2\pi n/\lambda \quad (2.14)$$

where λ remains the vacuum wavelength of the light. This relationship of the propagation constant to refractive index assumes a nonabsorbing medium ($\sigma = 0$). In the section below, the propagation of a plane wave will be expanded to include absorption.

2.1.3 Phasors and Wave Propagation in an Absorbing Medium

Although Equation (2.10) completely describes an arbitrary time-harmonic electromagnetic field, a more compact and convenient representation is possible based on the assumption that the frequency of the light does not change (quite a good assumption under normal optical conditions). A sinusoid can be related to a complex exponential by

$$E(P, t) = A(P) \cos(\omega t + \Phi(P)) = \text{Re}\{U(P)e^{-i\omega t}\} \quad (2.15)$$

where

$$U(P) = A(P)e^{-i\Phi(P)}$$

and $U(P)$ is called the *phasor* representation of the sinusoidal electric field $E(P, t)$. Notice that this phasor representation shows no time dependence. Study of the basic behavior of light has shown that the time dependence of the electric field typically does not change as light travels, interferes and interacts with matter. Thus, suppressing the time dependence and expressing the electric field as a phasor have become quite common in the mathematical analysis of optical systems. Fields that satisfy this assumption are called *time-harmonic fields*.

Separating out the time dependence for a time-harmonic wave makes evaluation of the time derivatives in Maxwell's equations straightforward. Using Equation (2.15) in Equation (2.5), and showing only the electric field for simplicity,

$$\nabla^2 E = \epsilon\mu \frac{\partial^2 E}{\partial t^2} + \mu\sigma \frac{\partial E}{\partial t} = -\omega^2 \epsilon\mu E - i\omega\mu\sigma E \quad (2.16)$$

Since every term in this equation contains the oscillatory $e^{-i\omega t}$, the time dependence can be suppressed by dividing this term out.

$$\nabla^2 U = -\omega^2 \epsilon\mu U - i\omega\mu\sigma U \quad (2.17)$$

Finally, this equation can be put into a standard form, known as the *Helmholtz equation*:

$$\nabla^2 U(P) + k^2 U(P) = 0 \quad (2.18)$$

where

$$k^2 = \omega^2 \epsilon\mu + i\omega\mu\sigma$$

The use of the symbol k in the Helmholtz equation is not coincidental. If $\sigma = 0$ (no absorption), k becomes the propagation constant defined in Equation (2.14). Thus, the

Helmholtz equation defines a complex propagation constant that accounts for absorption. Likewise, a complex refractive index can be defined.

$$\mathbf{n}^2 = \frac{k^2 \lambda^2}{4\pi^2} = \frac{\omega^2 \epsilon \mu \lambda^2}{4\pi^2} + i \frac{\omega \mu \sigma \lambda^2}{4\pi^2} = \mu c^2 \left(\epsilon + i \frac{\sigma}{\omega} \right) \quad (2.19)$$

Defining the complex refractive index in terms of real and imaginary parts,

$$\mathbf{n} = n + i\kappa \quad (2.20)$$

allows for the real and imaginary parts of the refractive index to be expressed in terms of the electrical and magnetic properties of the material as well as the frequency of oscillation:

$$\begin{aligned} n^2 &= \frac{1}{2} \mu c^2 \epsilon \left(\sqrt{1 + \left(\frac{\sigma}{\epsilon \omega} \right)^2} + 1 \right) \\ \kappa^2 &= \frac{1}{2} \mu c^2 \epsilon \left(\sqrt{1 + \left(\frac{\sigma}{\epsilon \omega} \right)^2} - 1 \right) \end{aligned} \quad (2.21)$$

Note that when $\sigma = 0$, $\kappa = 0$ and $\mathbf{n} = n$. For weakly absorbing materials, $\sigma/\epsilon\omega \ll 1$ and Equations (2.21) can be approximated as

$$\begin{aligned} n &\approx c \sqrt{\mu \epsilon} = \sqrt{\mu_r \epsilon_r} \\ \kappa &\approx \frac{1}{2} \left(\frac{c \sigma}{\omega} \right) \sqrt{\frac{\mu}{\epsilon}} = \frac{\lambda \sigma}{4\pi} \sqrt{\frac{\mu}{\epsilon}} \end{aligned} \quad (2.22)$$

Again assuming a nonmagnetic material, this weakly absorbing material will have an imaginary part that will be well approximated as

$$\frac{4\pi\kappa}{\lambda} \approx \frac{\sigma Z_0}{n} \quad \text{where} \quad Z_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 376.73 \Omega \quad (2.23)$$

and Z_0 is called the characteristic impedance of free space.

Since converting back to the time domain involves taking the real part of the phasor, the sign of the phase in $U(P)$ and in the time-dependent term could just as easily have been chosen to be positive rather than negative. This (arbitrary) sign convention is not consistent among authors and there is no absolute standard. The sign convention represented by Equation (2.15) is used by Goodman¹ and in other standard optics textbooks dealing with imaging. Using this convention, a plane wave traveling in the $+z$ direction [i.e. Equation (2.11)] would be written as

$$U(P) = A e^{ikz} \quad (2.24)$$

(where the phase offset δ has been ignored). This negative phase sign convention of Equation (2.15) is the most common for imaging applications. Unfortunately, many publications and textbooks in the area of thin film interference effects and coatings use the

positive phase sign convention. For this thin film sign convention, Equation (2.24) would represent a plane wave traveling in the $-z$ direction. Lithography requires both imaging calculations and thin film interference calculations. As such, both of these competing ‘standard’ sign conventions are often seen in lithography literature, though the negative sign convention of Equation (2.15) will be used throughout this book.

Equation (2.24) describes a plane wave traveling in the $+z$ direction. Generalizing to an arbitrary direction, let us first define the propagation vector $\mathbf{k}$ as the propagation constant multiplied by a unit vector pointing in the direction of propagation. Any point in space (x,y,z) is described by its position vector $\mathbf{r}$, a vector pointing from the origin to the point (x,y,z) . The phasor representation of the electric field of the plane wave is then given by

$$U(x,y,z) = Ae^{i\mathbf{k}\cdot\mathbf{r}} = Ae^{ik(\alpha x + \beta y + \gamma z)} \quad (2.25)$$

where α , β and γ are called the *direction cosines* of the propagation. As we shall see later in this chapter, the z -axis will coincide with the optical axis of an imaging system, so that the z -direction will be the main direction of light propagation. Thus, it will be convenient to define the direction of propagation using spherical coordinates about the z -axis. Letting θ be the direction $\mathbf{k}$ makes with the z -axis and ϕ be the angle that the k - z plane makes with the x -axis, the direction cosines become

$$\begin{aligned} \alpha &= \sin \theta \cos \phi \\ \beta &= \sin \theta \sin \phi \\ \gamma &= \sqrt{1 - \alpha^2 - \beta^2} = \cos \theta \end{aligned} \quad (2.26)$$

Another interesting and common solution to the wave equation is the spherical wave. For this wave, the wavefront is a sphere and the amplitude of the electric field is inversely proportional to the distance from the center of the sphere. In the equation below,

$$U(r) = \frac{Ae^{ikr}}{r} \quad (2.27)$$

the spherical waves are propagating outward from $r = 0$, so that the spherical wave can be thought of as emanating from a point source of light located at $r = 0$.

It is useful to derive some important properties of the plane wave. The time derivative of the field of a plane wave can be related to its spatial derivative with respect to the direction of propagation. Considering a plane wave traveling in the $+z$ direction, as given by Equation (2.11), but applied to the magnetic field:

$$\frac{\partial \mathbf{H}}{\partial t} = -\frac{\omega}{k} \frac{\partial \mathbf{H}}{\partial z} = -\frac{\omega}{k} \left(\frac{\partial H_x}{\partial z} \hat{x} + \frac{\partial H_y}{\partial z} \hat{y} \right) \quad (2.28)$$

Of course, an analogous equation applies to the electric field as well. The curl of the electric field for this plane wave becomes

$$\nabla \times \mathbf{E} = -\frac{\partial E_y}{\partial z} \hat{x} + \frac{\partial E_x}{\partial z} \hat{y} \quad (2.29)$$

These two equations can now be inserted into the second Maxwell equation, using the material property to relate B to H . The result, separated for each vector component, becomes

$$\begin{aligned} x\text{-component: } & -\frac{\partial E_y}{\partial z} - \frac{\omega\mu}{k} \frac{\partial H_x}{\partial z} = 0 \\ y\text{-component: } & +\frac{\partial E_x}{\partial z} - \frac{\omega\mu}{k} \frac{\partial H_y}{\partial z} = 0 \end{aligned} \quad (2.30)$$

Equations (2.30) can be integrated directly. Ignoring the constant of integration (any constant background electric or magnetic field will not concern us here),

$$E_y = -\frac{\omega\mu}{k} H_x = -\sqrt{\frac{\mu}{\epsilon}} H_x = -ZH_x, \quad E_x = \frac{\omega\mu}{k} H_y = \sqrt{\frac{\mu}{\epsilon}} H_y = ZH_y \quad (2.31)$$

where Z is called the characteristic impedance of the material. This gives a very important result. For a plane harmonic wave traveling through an isotropic material such that the material equations (2.2) apply, the electric and magnetic fields are related to each other by a multiplicative constant. Thus, knowing either one of these quantities is sufficient to determine the other. Moreover, it is the x -component of the magnetic field that is related to the y -component of the electric field, and vice versa. In other words, the electric field is always perpendicular to the magnetic field, and both are perpendicular to the direction of propagation. If Z is real, meaning that there is no absorption, the electric and magnetic fields will be in phase with each other.

2.1.4 Intensity and the Poynting Vector

When determining the intensity (also called irradiance[†]) transmitted into a material, it is very important to understand the exact definition of intensity. By definition, the *intensity* of light I is the magnitude of the (time averaged) Poynting vector, the energy per second crossing a unit area perpendicular to the direction of propagation of the light. The Poynting vector $\mathbf{S}$ is given by

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad (2.32)$$

where $\mathbf{E}$ is the electric field, and $\mathbf{H}$ is the magnetic field. (The derivation of this definition of intensity, not presented here, calculates the work performed by the electric and magnetic fields on the electrons surrounding the atoms of the medium.²)

For quasi-monochromatic time-harmonic fields, as represented by Equation (2.15), it will be convenient to express them as

$$\begin{aligned} \mathbf{E}(P, t) &= \text{Re}\{\mathbf{U}(P)e^{-i\omega t}\} = \frac{1}{2}[\mathbf{U}(P)e^{-i\omega t} + \mathbf{U}^*(P)e^{+i\omega t}] \\ \mathbf{H}(P, t) &= \text{Re}\{\mathbf{V}(P)e^{-i\omega t}\} = \frac{1}{2}[\mathbf{V}(P)e^{-i\omega t} + \mathbf{V}^*(P)e^{+i\omega t}] \end{aligned} \quad (2.33)$$

[†]By international standard, intensity is defined as power per unit solid angle while irradiance is power per unit area. However, the older nomenclature of intensity as power per unit area is still common (used in the classic textbook by Born and Wolf, for example), and will be used throughout this book.

where U^* is the complex conjugate of U . Using these equations, the cross product of the electric and magnetic fields becomes

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} = \frac{1}{4} [\mathbf{U} \times \mathbf{V} e^{-2i\omega t} + \mathbf{U}^* \times \mathbf{V} + \mathbf{U} \times \mathbf{V}^* + \mathbf{U}^* \times \mathbf{V}^* e^{+2i\omega t}] \quad (2.34)$$

Taking the time average of the Poynting vector is straightforward. For times much longer than the period of oscillation of the wave, the time harmonic terms will average to zero. Thus,

$$\langle \mathbf{S} \rangle = \frac{1}{4} [\mathbf{U}^* \times \mathbf{V} + \mathbf{U} \times \mathbf{V}^*] = \frac{1}{2} \text{Re}\{\mathbf{U} \times \mathbf{V}^*\} \quad (2.35)$$

To apply Equation (2.35) to a specific case, consider the monochromatic plane wave traveling in the z -direction. Remembering that electric and magnetic fields are perpendicular to each other and the direction of propagation, and using the results of Equation (2.31),

$$\mathbf{U} \times \mathbf{V}^* = (U_x V_y^* - U_y V_x^*) \hat{\mathbf{z}} = \frac{k}{\omega\mu} (U_x U_x^* + U_y U_y^*) \hat{\mathbf{z}} \quad (2.36)$$

Using the phasor expression for the plane wave, Equation (2.24),

$$\mathbf{U} \times \mathbf{V}^* = \frac{k}{\omega\mu} |\mathbf{U}|^2 \hat{\mathbf{z}} = \frac{k}{\omega\mu} |\mathbf{A}|^2 \hat{\mathbf{z}} \quad (2.37)$$

Thus, replacing $\text{Re}\{\omega/k\}$ with the speed of the wave from Equations (2.7) and (2.13), the intensity of the plane wave becomes

$$I = \langle \mathbf{S} \rangle = \frac{n}{2c\mu} |\mathbf{A}|^2 = \frac{1}{2} \sqrt{\frac{\epsilon}{\mu}} |\mathbf{A}|^2 \quad (2.38)$$

Since in general our media will be nonmagnetic (so that $\mu = \mu_0$), this final expression for intensity says that the intensity, within a multiplicative factor, is equal to the magnitude of the phasor electric field squared times the real part of the refractive index of the medium. Throughout this book, we will often have occasion to describe the intensity relative to some incoming wave, and so the multiplicative constant, $1/(2c\mu_0)$, will most often be ignored. And since throughout the rest of this book electric fields will always be expressed in phasor form, our general form for intensity will become

$$I = n |E|^2 \quad (2.39)$$

2.1.5 Intensity and Absorbed Electromagnetic Energy

The Poynting vector describes the flow of electromagnetic energy density. But the true purpose of knowing the intensity of light is to understand its impact on materials, which in our case means the impact of light on a photoresist. While the photochemistry of resists will be discussed in Chapters 5 and 6, suffice it to say here that the quantity of interest will be the electromagnetic energy absorbed by a material. Consider a monochromatic plane wave traveling through a uniform material of complex refractive index n . Separating the real and imaginary parts of the refractive index,

$$E = A e^{ikz} = A e^{i2\pi n z / \lambda} = A (e^{-2\pi \kappa z / \lambda}) e^{i2\pi n z / \lambda} \quad (2.40)$$

The intensity can then be written, from Equation (2.39),

$$I = n A^2 (e^{-4\pi \kappa z / \lambda}) = I_0 e^{-\alpha z} \quad (2.41)$$

where $I_0 = I(z = 0)$ and α is the absorption coefficient of the material, given by

$$\alpha = \frac{4\pi \kappa}{\lambda} \quad (2.42)$$

Equation (2.41) is the familiar *Lambert law of absorption*, empirically verified for homogeneous materials.

The imaginary part of the refractive index was related previously to the electrical and magnetic properties of the material in Equation (2.21). For a weakly absorbing material, such as a photoresist, the absorption coefficient becomes, from Equation (2.23)

$$\alpha \approx \frac{\sigma Z_0}{n} \quad \text{where} \quad Z_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 376.73 \Omega \quad (2.43)$$

Thus, absorption of electromagnetic energy results from the conductivity of the material at the frequency of the electromagnetic radiation. The electromagnetic wave gives energy to the electrons of the atoms of the material in proportion to its conductivity, in the form of absorbed photons of light.

2.2 Basic Imaging Theory

Using the basic understanding of light from the previous section, and especially the phasor representation of a plane wave, we are now ready to examine the behavior of an optical imaging system. Consider the generic projection system shown in Figure 2.1. It consists of a *light source*, a *condenser lens*, the *mask*, the *objective lens* and finally the resist-coated wafer. The combination of the light source and the condenser lens is called the *illumination system*. In optical design terms, a lens is a system of (possibly many) lens elements. Each lens element is an individual piece of glass (refractive element) or a mirror

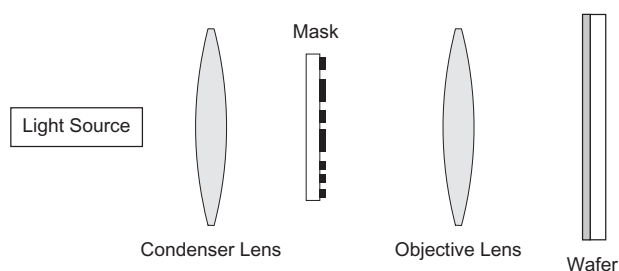


Figure 2.1 Block diagram of a generic projection imaging system

(reflective element) or other optical element. The purpose of the illumination system is to deliver light to the mask (and eventually into the objective lens) with sufficient intensity, the proper directionality and spectral characteristics, and adequate uniformity across the field (i.e. across the mask). The mask consists of a transparent substrate on which a pattern has been formed. This pattern changes the transmittance of the light and in its simplest form is just an opaque film. The light then passes through the clear areas of the mask and diffracts on its way to the objective lens. The purpose of the objective lens is to pick up a portion of the diffraction pattern and project an image onto the wafer which, one hopes, will resemble the mask pattern (or, more correctly, the desired pattern as expressed in the original design data).

2.2.1 Diffraction

The first and most basic phenomenon occurring in projection imaging is the diffraction of light. Diffraction is usually thought of as the bending of light as it passes through an aperture, which is certainly an appropriate description for diffraction by a lithographic mask. More correctly, diffraction theory simply describes how light propagates, including the effects of the surroundings (boundaries). Maxwell's equations describe how electromagnetic waves propagate, but result in partial differential equations of vector quantities which, for general boundary conditions, are extremely difficult to solve without the aid of a powerful computer and sophisticated numerical algorithms. A simpler approach is applicable to the propagation of light within a homogenous medium (for example, describing how light travels from the photomask to the objective lens of our imaging system). This simple approach leads to the development of *diffraction integrals* derived from the Helmholtz equation (2.18). While the derivation of these diffraction integrals will not be presented here, some historical background will help to put their usefulness in perspective.

A simple interpretation of the physical principle behind diffraction is best captured by *Huygens' principle*. First described by the famous Dutch scientist Christian Huygens in his 1690 *Treatise on Light*, any wavefront can be thought of as a collection of radiating point sources (Figure 2.2). The new wavefront at some later time can be constructed by summing up the wavefronts from all of the radiated spherical waves. As Figure 2.2 shows, one result of these self-luminous wavefront points is the apparent bending of light around an opaque edge – an example of diffraction. In 1818, Augustin-Jean Fresnel formed a



Figure 2.2 Pictorial representation of Huygens' principle, where any wavefront can be thought of as a collection of point sources radiating spherical waves

mathematical theory of diffraction by turning this summation into an integral and accounting for the phase of the light when adding together the propagating spherical waves. He also included an empirical obliquity factor to keep these radiating spherical waves from propagating backwards as readily as forward. This scalar diffraction theory was put on a more rigorous footing by the Prussian mathematician and physicist Gustav Kirchhoff in 1882, who required the diffracting waves to satisfy the Helmholtz equation and conservation of energy, thus deriving Fresnel's previously empirical obliquity term. Fresnel's diffraction integral is in fact a simplification of Kirchhoff's formulation for the case when the distance away from the diffracting plane (that is, the distance from the mask to the objective lens) is much greater than the wavelength of light. Finally, if the distance to the objective lens is very large, or if the mask is illuminated by a spherical wave which converges to a point at the entrance to the objective lens, Fresnel diffraction simplifies to *Fraunhofer diffraction* (named for Joseph von Fraunhofer, who used a lens to create the far-field diffraction pattern). Comparison of these different diffraction regions is given in Figure 2.3.

In order to establish a mathematical description of diffraction by a mask, we must first describe the electric field transmittance of a mask pattern $t_m(x,y)$, where the mask is in the x - y plane and $t_m(x,y)$ has in general a magnitude, phase and vector direction. An exact calculation of the mask transmittance requires a solution to Maxwell's equations for the given materials and physical structure of the mask and for a specific incident light wave. As we shall see in the sections below, any arbitrary incident illumination can be conveniently broken up into a summation of plane waves of different polarizations and propagation directions. Thus, the transmittance of different polarizations of incident light as a function of the incident angle of light is required.

In some circumstances, however, a simpler approach will prove adequate. For cases where the feature sizes on the mask are larger than about 2λ and where the topography on the mask (for example, the thickness of a chrome absorber) is less than about $\lambda/2$, it is possible to ignore diffraction by the topography of the mask and simply assume that the mask transmittance is a perfect shadow of the mask. In other words, the transmittance directly below a specific feature on the mask is assumed to be the same as the transmittance under an infinitely large area of that material. At an edge between two materials, the transmittance of the boundary is assumed to be a step function between the transmittances of the two materials. This assumed transmittance function is called the *Kirchhoff*

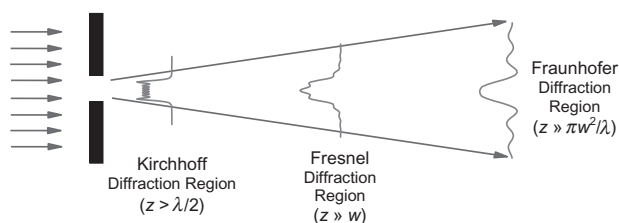


Figure 2.3 Comparison of the diffraction 'regions' where various approximations become accurate. Diffraction is for a slit of width w illuminated by light of wavelength λ , and z is the distance away from the mask

boundary condition. Note that under this assumption, the transmittance for each polarization is assumed to be the same, so that incident polarization can be ignored when employing the Kirchhoff boundary condition. Also, the effect of incident angle can be treated very simply in the case of the Kirchhoff boundary condition, as will be discussed in section 2.3.

For a simple chrome-glass mask, the mask E-field transmittance for a normally incident plane wave of illumination under the Kirchhoff boundary condition assumption is binary: $t_m(x,y)$ is 1 under the glass and 0 under the chrome. Let the $x'-y'$ plane be the diffraction plane, that is, the entrance to the objective lens, and let z be the distance from the mask to this diffraction plane. We will also assume monochromatic light of wavelength λ and that the entire system is in a medium of refractive index n . Defining the *spatial frequencies* of the diffraction pattern (which are simply scaled coordinates in the $x'-y'$ plane) as $f_x = nx'/(z\lambda)$ and $f_y = ny'/(z\lambda)$, the electric field of our diffraction pattern, $T_m(f_x, f_y)$, is given by the *Fraunhofer diffraction integral*:

$$T_m(f_x, f_y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} E_i(x,y) t_m(x,y) e^{-2\pi i(f_x x + f_y y)} dx dy \quad (2.44)$$

where E_i is the electric field incident on the mask (and is just 1 for our unit amplitude, normally incident plane wave). For many scientists and engineers, this equation should be quite familiar – it is simply a *Fourier transform*. Thus, the diffraction pattern (i.e. the electric field distribution as it enters the objective lens) is just the Fourier transform of the mask pattern transmittance (the electric field directly under the mask). This is the principle behind an extremely useful approach to imaging called Fourier optics (for more information, consult Goodman's classic textbook¹).

Figure 2.4 shows two mask patterns – one an isolated space, the other a series of equal lines and spaces – both infinitely long in the y -direction (the direction out of the page). The resulting one-dimensional mask field transmittance functions, $t_m(x)$, look like a square pulse and a square wave, respectively. The Fourier transforms for normally incident plane wave illumination ($E_i = 1$) are easily computed directly from Equation (2.44) or found in tables or textbooks (see Problem 2.4) and are also shown in Figure 2.4. The isolated space

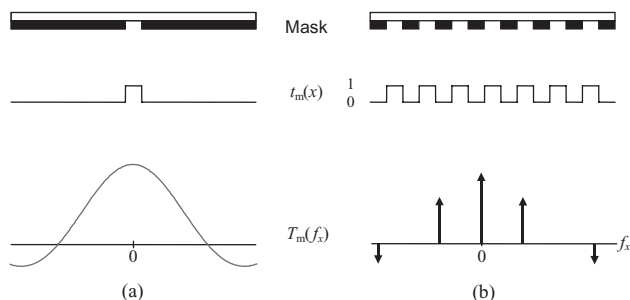


Figure 2.4 Two typical mask patterns, (a) an isolated space and (b) an array of equal lines and spaces, and the resulting Fraunhofer diffraction patterns assuming normally incident plane wave illumination. Both t_m and T_m represent electric fields

gives rise to a *sinc* function ($\sin(x)/x$) diffraction pattern, and the equal lines and spaces yield discrete *diffraction orders*:

$$\begin{aligned} \text{Isolated space: } T_m(f_x) &= \frac{\sin(\pi w f_x)}{\pi f_x} \\ \text{Dense space: } T_m(f_x) &= \frac{1}{p} \sum_{j=-\infty}^{\infty} \frac{\sin(\pi w f_x)}{\pi f_x} \delta\left(f_x - \frac{j}{p}\right) \end{aligned} \quad (2.45)$$

where δ is the Dirac delta function, w is the spacewidth and p is the pitch (the linewidth plus the spacewidth). The delta function is the mathematical representation of a point of light. (As a review, the properties of the delta function are discussed in Appendix C.)

It is interesting to consider how different these two diffraction patterns are. The isolated space results in a continuous distribution of energy in the form of a sinc function. The dense array of spaces produces discrete points of light, whose brightness follows the sinc function envelop (that is, the diffraction pattern of just one of its spaces). To visualize how this happens, consider the intermediate cases of two or three spaces of equal width, separated by lines of the same width. Figure 2.5 shows the intensity of the diffraction patterns (electric field amplitude squared) for these cases. For the multiple space cases, diffraction patterns from adjacent spaces interfere with each other, producing narrow interference fringes. The greater the number of spaces, the greater the interference is and thus the narrower the peaks. As the number of spaces in the array approaches infinity, the fringes become infinitely narrow points of light.

Let's take a closer look at the diffraction pattern for equal lines and spaces. At the center of the objective lens entrance ($f_x = 0$), the diffraction pattern has a bright spot called the *zero order*. The zero order is the light that passes through the mask and is not bent. To either side of the zero order are two peaks called the *first diffraction orders*. These peaks occur at spatial frequencies of $\pm 1/p$, where p is the pitch of the mask pattern (linewidth plus spacewidth). Since the position of these diffraction orders relative to the zero

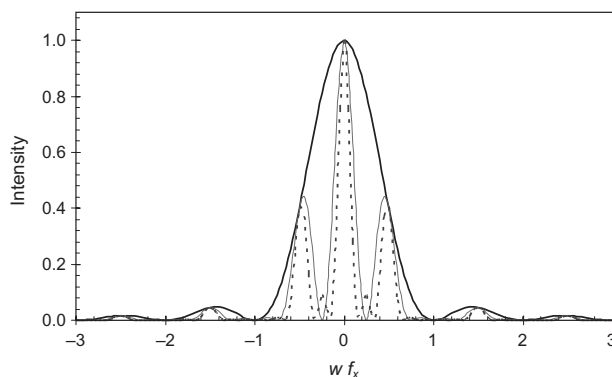


Figure 2.5 Magnitude of the diffraction pattern squared (intensity) for a single space (thick solid line), two spaces (thin solid line) and three spaces (dashed lines) of width w . For the multiple-feature cases, the linewidth is also equal to w

order depends on the mask pitch, their relative positions contain information about the pitch. It is this information that the objective lens will use to reproduce the image of the mask. In fact, in order for the objective lens to form a true image of a pattern of lines and spaces, at least two diffraction orders must be captured by the objective lens and used to form the image. In addition to the first order, there can be many higher orders, with the j th order occurring at a spatial frequency of j/p .

While in principle the number of diffraction orders (the range of values of j) is infinite, only some of these orders are capable of propagating from the mask to the lens. If a wave has a spatial frequency such that

$$\sqrt{f_x^2 + f_y^2} > \frac{1}{\lambda} \quad (2.46)$$

these waves are said to *evanescent*. This condition is equivalent to a wave whose direction cosines have $\alpha^2 + \beta^2 > 1$ so that the direction cosine γ is imaginary. The amplitudes of these evanescent waves decay exponentially as

$$e^{-\zeta z} \quad \text{where} \quad \zeta = 2\pi n \sqrt{f_x^2 + f_y^2 - \frac{1}{\lambda^2}} \quad (2.47)$$

Thus, for practical purposes, diffraction patterns need only be considered out to spatial frequencies less than the onset of evanescent waves.

Summarizing, given a mask in the x - y plane described by its electric field transmission $t_m(x,y)$ multiplied by the incident electric field $E_i(x,y)$, the electric field T_m as it enters the objective lens (the x' - y' plane) is given by

$$T_m(f_x, f_y) = \mathcal{F}\{E_i(x,y)t_m(x,y)\} \quad (2.48)$$

where the symbol $\mathcal{F}$ represents the Fourier transform and f_x and f_y are the spatial frequencies and are simply scaled coordinates in the x' - y' diffraction plane. Often, $t_m(x,y)$ is approximated as the shadow transmittance of the mask using the Kirchhoff boundary condition. For normally incident plane wave illumination, $E_i = 1$.

2.2.2 Fourier Transform Pairs

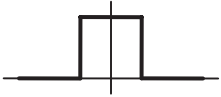
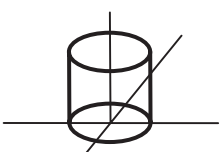
Some important diffraction patterns can now be calculated. Table 2.1 shows several common 1D functions of some interest to lithographers, as well as their Fourier transforms. These transform pairs can be applied to different problems with the addition of two important theorems. If $\mathcal{F}\{g(x,y)\} = G(f_x, f_y)$, the *shift theorem* of the Fourier transform says

$$\mathcal{F}\{g(x-a, y-b)\} = G(f_x, f_y) e^{-i2\pi(f_x a + f_y b)} \quad (2.49)$$

where a and b are constants. In other words, a shift in position of the mask produces a linear phase variation across the diffraction pattern. The *similarity theorem* of the Fourier transform is

$$\mathcal{F}\{g(ax, by)\} = \frac{1}{|ab|} G\left(\frac{f_x}{a}, \frac{f_y}{b}\right) \quad (2.50)$$

Table 2.1 1D Fourier transform pairs useful in lithography

$g(x)$	Graph of $g(x)$	$G(f_x)$
$rect(x) = \begin{cases} 1, & x < 0.5 \\ 0, & x > 0.5 \end{cases}$		$\frac{\sin(\pi f_x)}{\pi f_x}$
$step(x) = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases}$		$\frac{1}{2} \delta(f_x) - \frac{i}{2\pi f_x}$
Delta function $\delta(x)$		1
$comb(x) = \sum_{j=-\infty}^{\infty} \delta(x - j)$		$\sum_{j=-\infty}^{\infty} \delta(f_x - j)$
$\cos(\pi x)$		$\frac{1}{2} \delta\left(f_x + \frac{1}{2}\right) + \frac{1}{2} \delta\left(f_x - \frac{1}{2}\right)$
$\sin(\pi x)$		$\frac{i}{2} \delta\left(f_x + \frac{1}{2}\right) - \frac{i}{2} \delta\left(f_x - \frac{1}{2}\right)$
Gaussian $e^{-\pi x^2}$		$e^{-\pi f_x^2}$
$circ(r) = \begin{cases} 1, & r < 1 \\ 0, & r > 1 \end{cases}$ $r = \sqrt{x^2 + y^2}$		$\frac{J_1(2\pi\rho)}{\pi\rho}$ $\rho = \sqrt{f_x^2 + f_y^2}$

This theorem shows that making a pattern smaller ($a, b < 1$) spreads the diffraction pattern to higher spatial frequencies. Finally, by recognizing that the Fourier transform is a linear operation (see Problem 2.7), superposition of known Fourier transform pairs can be used to build more complicated functions.

Another important general transform that is extremely useful in lithography is the generalized repeating pattern. Consider a two-dimensional pattern that repeats in x and y with pitches p_x and p_y , respectively. Let $t_r(x, y)$ be the transmittance of a single repeating unit such that the transmittance is nonzero only in the range $-p_x/2 < x < p_x/2$ and $-p_y/2 < y < p_y/2$. The total transmittance of the mask is then

$$t_m(x,y) = \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} t_r(x + jp_x, y + kp_y) = t_r(x,y) \otimes \text{comb}\left(\frac{x}{p_x}, \frac{y}{p_y}\right) \quad (2.51)$$

where the ‘ $\otimes$ ’ symbol denotes convolution. Letting $T_r(f_x, f_y)$ be the Fourier transform of the isolated repeating unit transmittance, the Fourier transform of Equation (2.51) gives

$$T_m(f_x, f_y) = \frac{1}{p_x p_y} \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} T_r(f_x, f_y) \delta\left(f_x - \frac{j}{p_x}, f_y - \frac{k}{p_y}\right) \quad (2.52)$$

Thus, any repeating pattern produces a diffraction pattern made up of delta functions (diffraction orders). The amplitudes of the diffraction orders are given by the Fourier transform of the isolated repeating unit mask pattern.

2.2.3 Imaging Lens

We are now ready to describe what happens next and follow the diffracted light as it enters the objective lens. In general, the diffraction pattern extends throughout the $x'-y'$ plane. However, the objective lens, being only of finite size, cannot collect all of the light in the diffraction pattern. Typically, lenses used in microlithography are circularly symmetric and the entrance to the objective lens can be thought of as a circular aperture. Only those portions of the mask diffraction pattern that fall inside the aperture of the objective lens go on to form the image. Of course, we can describe the size of the lens aperture by its radius, but a more common and useful description is to define the maximum angle of diffracted light that can enter the lens. Consider the geometry shown in Figure 2.6. Light passing through the mask is diffracted at various angles. Given a lens of a certain size placed at a certain distance from the mask, there is some maximum angle of diffraction, $\theta_{\max}$, for which the diffracted light just makes it into the lens. Light emerging from the mask at larger angles misses the lens and is not used in forming the image. The most convenient way to describe the size of the lens aperture is by its *numerical aperture*,

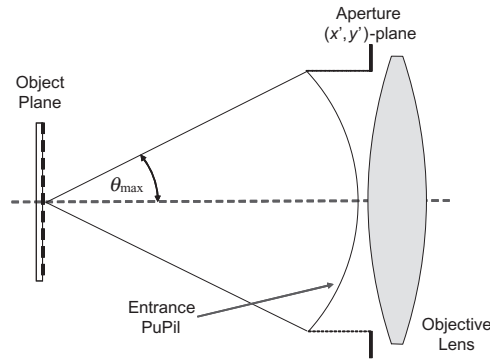


Figure 2.6 The numerical aperture is defined as $NA = n \sin \theta_{\max}$ where $\theta_{\max}$ is the maximum half-angle of the diffracted light that can enter the objective lens, and n is the refractive index of the medium between the mask and the lens

defined as the sine of the maximum half-angle of diffracted light that can enter the lens times the index of refraction of the surrounding medium, n .

$$NA = n \sin \theta_{\max} \quad (2.53)$$

Besides the diffraction plane, it is important to define the *entrance pupil* of the objective lens. Despite the simplicity of the single-lens diagram in Figure 2.6, the diffraction plane (also called the *pupil plane* or *aperture plane*) is actually buried inside the lens in most cases. The entrance pupil (or opening) is spherical in shape and is the image of the diffraction plane as seen from the entrance of the lens. The diffraction plane can be constructed simply as the geometric projection of the spherical entrance pupil onto a plane. (Likewise, the *exit pupil* is the image of the aperture plane as viewed from the exit side of the lens.) Thus, one can easily express the spatial frequency in terms of the angle of diffraction:

$$f_x = \frac{nx'}{\lambda z} = \frac{n \sin \theta_x}{\lambda} \quad (2.54)$$

with, of course, an equivalent expression for the y -direction. Alternately, the spatial frequencies can be related to the direction cosines as defined in Equation (2.25):

$$f_x = \frac{n\alpha}{\lambda}, \quad f_y = \frac{n\beta}{\lambda} \quad (2.55)$$

Using this description, the maximum spatial frequency that can enter the objective lens is given by NA/λ .

Clearly, the numerical aperture is going to be quite important. A large numerical aperture means that a larger portion of the diffraction pattern is captured by the objective lens. For a small numerical aperture, much more of the diffracted light is lost. In fact, we can use this viewpoint to define *resolution*, at least from the limited perspective of image formation. Consider the simple case of a mask pattern of equal lines and spaces. As we have seen, the resulting diffraction pattern is a series of discrete diffraction orders. In order to produce an image that even remotely resembles the original mask pattern, it is necessary for the objective lens to capture the zero order and at least one higher diffraction order. If the light illuminating the mask is a normally incident plane wave (the only type of illumination we have discussed so far), the diffraction pattern will be centered in the objective lens (more on this topic will be covered in section 2.3.1). Since the positions of the ± 1 st diffraction orders in frequency space are given by $\pm 1/p$, the requirement that a lens must capture these diffraction orders to form an image puts a lower limit on the pitch that can be imaged. Thus, the smallest pitch ($p_{\min}$) that still produces an image will put the first diffraction order at the outer edge of the objective lens:

$$\frac{1}{p_{\min}} = \frac{NA}{\lambda} \quad (2.56)$$

If we let R represent the resolution element (the linewidth or the spacewidth) of our equal line/space pattern, the resolution will be given by

$$R = 0.5 \frac{\lambda}{NA} \quad (2.57)$$

This classic equation is often called the *theoretical resolution* of an imaging system. Note that several very specific assumptions were made in deriving this resolution equation: a mask pattern of equal lines and spaces was used, and the illumination was a single wavelength normally incident plane wave (called coherent illumination). In reality, this equation defines the *pitch resolution*, the smallest pitch that can be imaged, for normally incident plane waves. For other features, for example an isolated line, there is no clear resolution cutoff. For other types of illumination, this cutoff will change. As a result, this approximate resolution expression is often generalized as

$$R = k_1 \frac{\lambda}{NA} \quad (2.58)$$

Such resolution equations are best interpreted as scaling equations, with k_1 as the scaled resolution. See Chapter 10 for a more thorough discussion of resolution.

2.2.4 Forming an Image

To proceed further, we must now describe how the lens affects the light entering it. Obviously, we would like the image to resemble the mask pattern. Since diffraction gives the Fourier transform of the mask, if the lens could give the *inverse* Fourier transform of the diffraction pattern, the resulting image would resemble the mask pattern. In fact, lenses are designed to behave precisely in this way. We can define an ideal imaging lens as one that produces an image at the focal plane that is identically equal to the Fourier transform of the light distribution entering the lens. A property of the Fourier transform is

$$\mathcal{F}\{\mathcal{F}\{g(x,y)\}\} = g(-x, -y) \quad (2.59)$$

Thus, one Fourier transform (diffraction) followed by a second Fourier transform (focusing by the imaging lens) produces an image that is a 180° rotated version of the original. It is common to define an image coordinate system that is rotated by 180° so that this rotation can be ignored. In the rotated image coordinate system, the focusing behavior of the imaging lens is described as a simple inverse Fourier transform.

It is the goal of lens designers and manufacturers to create lenses as close as possible to this ideal. Does an ideal lens produce a perfect image? No. Because of the finite size of the lens aperture, only a portion of the diffraction pattern enters the lens. Thus, even an ideal lens cannot produce a perfect image unless the lens is infinitely big. Since in the case of an ideal lens the image is limited only by the diffracted light that does not make it through the lens, we call such an ideal system *diffraction limited*.

In order to write our final equation for the formation of an image, let us define the objective lens *pupil function* P (a pupil is just another name for an aperture) as the transmittance of the lens from the entrance pupil to the exit pupil. The pupil function of an ideal lens simply describes what portion of light makes it through the lens: it is one inside the aperture and zero outside.

$$P(f_x, f_y) = \begin{cases} 1, & \sqrt{f_x^2 + f_y^2} < NA/\lambda \\ 0, & \sqrt{f_x^2 + f_y^2} > NA/\lambda \end{cases} \quad (2.60)$$

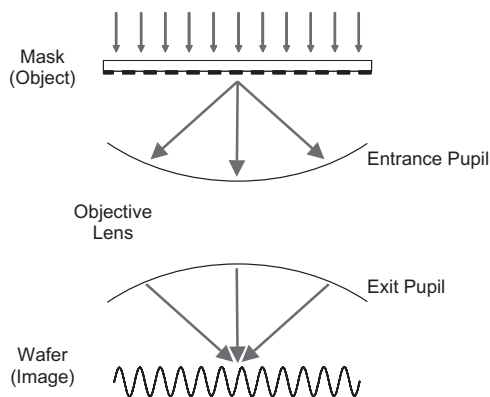


Figure 2.7 Formation of an aerial image: a pattern of lines and spaces produces a diffraction pattern of discrete diffraction orders (in this case, three orders are captured by the lens). The lens turns these diffraction orders into plane waves projected onto the wafer, which interfere to form the image

Thus, the product of the pupil function and the diffraction pattern describes the light transmitted through the objective lens, that is, the light at the exit pupil. Combining this with our description of how a lens behaves gives us our final expression for the electric field at the image plane (that is, at the wafer):

$$E(x,y) = \mathcal{F}^{-1}\{T_m(f_x, f_y)P(f_x, f_y)\} \quad (2.61)$$

where the symbol $\mathcal{F}^{-1}$ represents the inverse Fourier transform. The *aerial image* is defined as the intensity distribution in air at the wafer plane and is simply the square of the magnitude of the electric field (see section 2.1.3).

Consider the full imaging process (Figure 2.7). First, light passing through the mask is diffracted. The diffraction pattern can be described as the Fourier transform of the electric field at the bottom of the mask. Since the objective lens is of finite size, only a portion of the diffraction pattern actually enters the lens. The numerical aperture defines the maximum angle of diffracted light that enters the lens and the pupil function is used to mathematically describe this behavior. Finally, the effect of the lens is to take the inverse Fourier transform of the light exiting the lens to give an image which (one hopes) resembles the mask pattern. If the lens is ideal, the quality of the resulting image is only limited by the amount of the diffraction pattern collected. This type of imaging system is called diffraction limited.

2.2.5 Imaging Example: Dense Array of Lines and Spaces

Again referring to the simplified case of an infinite array of lines and spaces, the diffraction pattern is made up of discrete diffraction orders, points of light represented mathematically as delta functions. The image of such a pattern is calculated as the inverse Fourier transform of the diffraction orders that make it through the lens (those with spatial frequencies less than NA/λ). The inverse Fourier transform of a delta function is just 1, so that a point source in the pupil plane of the lens produces a plane wave at the wafer

plane. Letting N be the largest diffraction order that passes through the lens, and using the dense space diffraction pattern of Equation (2.45),

$$E(x) = \int_{-\infty}^{\infty} \frac{1}{p} \sum_{j=-N}^N \frac{\sin(\pi w f_x)}{\pi f_x} \delta\left(f_x - \frac{j}{p}\right) e^{i2\pi f_x x} df_x \quad (2.62)$$

$$E(x) = \sum_{j=-N}^N a_j e^{i2\pi jx/p} \quad \text{where} \quad a_j = \frac{\sin(j\pi w/p)}{j\pi}$$

The term $e^{i2\pi jx/p}$ represents the electric field of a unit-amplitude plane wave spread out over the focal plane, whose incident angle when striking the focal plane is given by the angle of the j th diffraction order ($\sin \theta = j\lambda/p$). The amplitude of this plane wave is a_j , the amplitude of the j th diffraction order. The electric field image at the wafer plane is the sum (the interference) of the various plane waves reaching the wafer. This interpretation of the line/space image result is extremely useful. As Equation (2.52) shows, any repeating pattern on the mask will produce a diffraction pattern made up of delta functions (point sources) called diffraction orders. Each diffraction order in turn produces one plane wave striking the wafer. The summation (interference) of all of the plane waves produces the image. This is sometimes called *plane wave decomposition*, since the image is decomposed into plane wave basis functions.

Since the diffraction pattern for this case is centered in the entrance pupil, Equation (2.62) can be simplified using Euler's theorem.

$$E(x) = a_0 + 2 \sum_{j=1}^N a_j \cos(2\pi jx/p) \quad (2.63)$$

The zero order, given by a_0 , provides a DC offset to the electric field. Each pair of diffraction orders, ± 1 , ± 2 , etc., provides a cosine at harmonics of the pitch p . For the case of equal lines and spaces, the diffraction order amplitudes become

$$a_j = \frac{\sin(j\pi/2)}{j\pi} = \begin{cases} \frac{1}{2}, & j = 0 \\ 0, & j = \text{even} \\ \frac{1}{|j|\pi}, & j = \pm 1, \pm 5, \pm 9, \dots \\ -\frac{1}{|j|\pi}, & j = \pm 3, \pm 7, \pm 11, \dots \end{cases} \quad (2.64)$$

A graph of this diffraction pattern for ± 3 orders is given in Figure 2.8.

Consider the case of lines and spaces where $N = 1$ (that is, the lens captures the zero order and the ± 1 orders). The electric field image is then

$$E(x) = a_0 + 2a_1 \cos(2\pi x/p) = \frac{1}{2} + \frac{2}{\pi} \cos(2\pi x/p) \quad (2.65)$$

and the intensity image is

$$I(x) = |E(x)|^2 = a_0^2 + 4a_0a_1 \cos(2\pi x/p) + 4a_1^2 \cos^2(2\pi x/p) \quad (2.66)$$

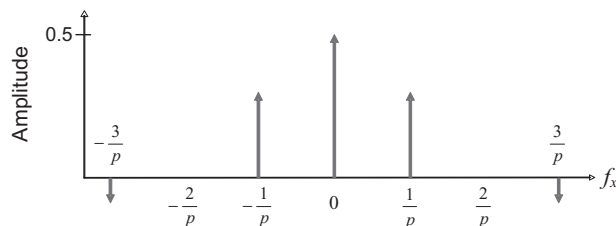


Figure 2.8 Graph of the diffraction pattern for equal lines and spaces plotted out to ± 3 rd diffraction orders. For graphing purposes, the delta function is plotted as an arrow, the height equal to the amplitude multiplier of the delta function

For equal lines and spaces, the image becomes

$$I(x) = \frac{1}{4} + \frac{2}{\pi} \cos(2\pi x/p) + \frac{4}{\pi^2} \cos^2(2\pi x/p) \quad (2.67)$$

or

$$I(x) = \frac{1}{4} + \frac{2}{\pi^2} + \frac{2}{\pi} \cos(2\pi x/p) + \frac{2}{\pi^2} \cos(4\pi x/p) \quad (2.68)$$

It is useful to interpret each term in Equation (2.67) physically. The zero order contributes a DC bias to the final image, seen as the first term on the right-hand side of Equation (2.67). The second term represents the interference of the zero order with the first orders and contains the main cosine function that determines the overall shape of the aerial image. The final cosine-squared term represents the interference of the $+1$ order with the -1 order. Since the distance between these orders is $2/p$ in spatial frequency space, this term contributes a frequency-doubled sinusoid to the image.

As a note, the image calculations above perform a *scalar* addition of the electric fields of the zero and first orders. That is, the vector directions of the electric fields of each plane wave interfering at the wafer plane are assumed to be parallel. For the case where this is not true, a full vector treatment of the image calculation, as given in Chapter 3, is required.

The number of diffraction orders captured by the objective lens determines the quality and characteristics of the resulting aerial image. If the numerical aperture of the lens is increased (or if the wavelength is decreased), increasing numbers of diffracted orders can be captured and used to form the image. Each added diffraction order contains information that improves the quality of the aerial image (Figure 2.9).

2.2.6 Imaging Example: Isolated Space

Consider now the image of an isolated space under coherent illumination. Taking the inverse Fourier transform of the portion of the diffraction pattern [from Equation (2.45)] that makes it through the lens,

$$E(x) = \int_{-NA/\lambda}^{NA/\lambda} \frac{\sin(\pi w f_x)}{\pi f_x} e^{i2\pi f_x x} df_x \quad (2.69)$$

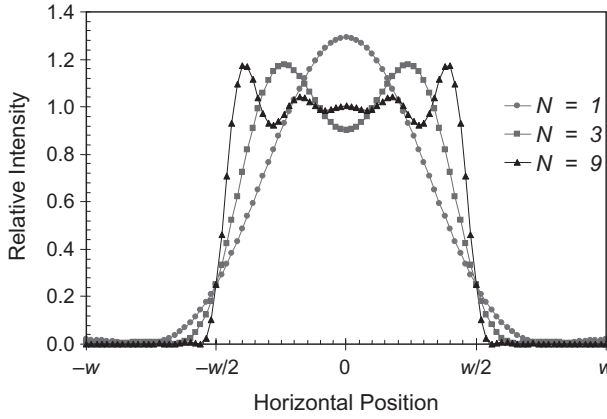


Figure 2.9 Aerial images for a pattern of equal lines and spaces of width w as a function of the number of diffraction orders captured by the objective lens (coherent illumination). N is the maximum diffraction order number captured by the lens

Breaking up the complex exponential into a sine and cosine using Euler's identity, any odd part of the function will integrate to zero. Thus, keeping only the even part,

$$E(x) = 2 \int_0^{NA/\lambda} \frac{\sin(\pi w f_x) \cos(2\pi x f_x)}{\pi f_x} df_x \quad (2.70)$$

Using a trigonometric identity,

$$E(x) = \int_0^{NA/\lambda} \frac{\sin(2\pi f_x [x + w/2]) - \sin(2\pi f_x [x - w/2])}{\pi f_x} df_x \quad (2.71)$$

This integral can be expressed in terms of the commonly tabulated Sine integral (Si) to give the final image intensity:

$$I(x) = \frac{1}{\pi^2} \left[\text{Si} \left(\frac{2\pi NA}{\lambda} (x + w/2) \right) - \text{Si} \left(\frac{2\pi NA}{\lambda} (x - w/2) \right) \right]^2 \quad (2.72)$$

where

$$\text{Si}(\theta) = \int_0^\theta \frac{\sin z}{z} dz$$

2.2.7 The Point Spread Function

A common method of characterizing the resolving capability of an imaging system is to consider the smallest possible contact hole that can be printed. (The term 'contact' hole refers to the use of this small hole in ICs for making electrical contact between two metal layers separated by an insulator.) Consider a mask pattern of an isolated rectangular hole in a chrome (totally dark) background. Now let that hole shrink to an infinitesimal pinhole.

As the hole shrinks, the intensity of light reaching the wafer becomes infinitesimally small, but we shall normalize out this effect and consider only the shape of the resulting aerial image. Thus, as the contact hole shrinks to a pinhole, the transmittance of the photomask becomes a delta function. The diffraction pattern, then, is just 1, a plane wave uniformly filling the pupil of the objective lens.

For uniform filling of the entrance pupil, the resulting image is the inverse Fourier transform of the pupil function. The result for an ideal circular aperture is

$$E(x,y) = \mathcal{F}^{-1}\{P(f_x, f_y)\} = \frac{J_1(2\pi\rho)}{\pi\rho} \quad (2.73)$$

where J_1 is the Bessel function of the first kind, order one, and ρ is the radial distance from the center of the image normalized by multiplying by NA/λ . The intensity aerial image of a pinhole, when normalized in this way, is called the *point spread function* (PSF) of the optical system.

$$PSF_{\text{ideal}} = \left| \frac{J_1(2\pi\rho)}{\pi\rho} \right|^2 \quad (2.74)$$

The PSF (also called the *Airy disk*, after George Biddle Airy) is a widely used metric of imaging quality for optical system design and manufacture and is commonly calculated for lens designs and measured on fabricated lenses using special bench-top equipment. Figure 2.10 shows a graph of the ideal PSF.

How wide is the PSF? For large contact holes, the normalized intensity at a position corresponding to the mask edge (that is, at the desired contact hole width) is about 0.25–0.3. If we use this intensity range to measure the width of the PSF, the result is a contact hole between 0.66 and 0.70 λ/NA wide. Thus, this width represents the smallest possible contact hole that could be imaged with a conventional chrome on glass (i.e. not phase-shifted) mask. If the contact hole size on the mask approaches or is made smaller than

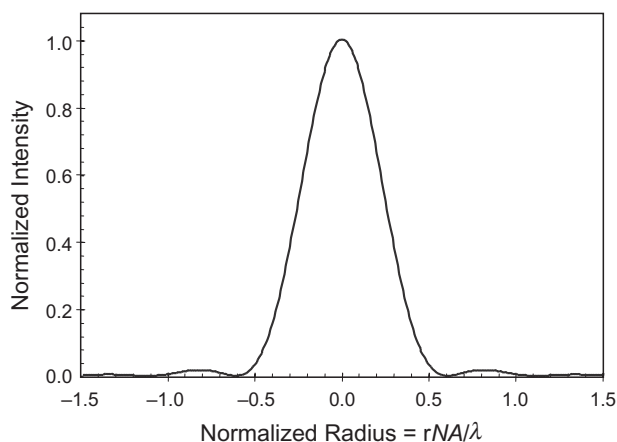


Figure 2.10 Ideal point spread function (PSF), the normalized image of an infinitely small contact hole. The radial position is normalized by multiplying by NA/λ

this value, the printed image is controlled by the PSF, not by the dimensions of the mask! Making the contact size on the mask smaller only reduces the intensity of the image peak. Thus, this width of the PSF, the ultimate resolution of a chrome on glass (COG) contact hole, is a *natural resolution* of the imaging system.

The PSF can also be interpreted using linear systems theory. For the coherent illumination discussed so far, the Fourier transform of the electric field image is equal to the pupil function multiplied by the Fourier transform of the mask transmittance.

$$\mathcal{F}\{E(x,y)\} = T_m(f_x, f_y)P(f_x, f_y) \quad (2.75)$$

Thus, the electric field image is linear in Fourier space, with the pupil function serving the role of the system transfer function. In real space, applying the convolution theorem of the Fourier transform to Equation (2.75) gives

$$E(x,y) = t_m(x,y) \otimes \mathcal{F}^{-1}\{P(f_x, f_y)\} \quad (2.76)$$

The inverse Fourier transform of the pupil function, as we saw above, is just the electric field PSF. Thus, for coherent illumination, the electric field image is the electric field mask transmittance convolved with the electric field PSF.

In linear systems theory, the electric field PSF would be called the *impulse response* of the imaging system. An infinitely small pinhole on the mask produces a delta function or ‘impulse’ of light as the object, so that the PSF, being the image of that impulse object, is called the impulse response of the imaging system. For fields, the Green’s function plays the equivalent role of the impulse response and the PSF will sometimes be referred to as a Green’s function.

2.2.8 Reduction Imaging

The above discussion makes a useful simplifying assumption – that the imaging system is 1×, or unit magnification, so that the dimensions of the mask patterns are nominally the same as those of the image being produced. In reality, most imaging tools used in lithography are 4× to 10× reduction systems (with 4× the most common), so that the mask is made 4× to 10× bigger than the desired pattern sizes on the wafer. Fortunately, this reduction can be accounted for by a simple scaling relationship so that imaging with reduction is treated mathematically the same as for a 1× system. For a reduction ratio R , all lateral (x and y) dimensions at the wafer are equal to the mask dimensions divided by R . Additionally, the sine of the angles of diffraction are R times greater on the wafer side (exit side) of the lens compared to the mask side (entrance side). Thus, the common numerical aperture used to describe the size of the lens is more properly defined as the exit pupil numerical aperture, the entrance pupil NA being R times smaller (Figure 2.11). Likewise, spatial frequencies on the wafer side are R times larger than spatial frequencies on the mask side (a property of any good imaging system known as the *Abbe sine condition*):

$$n_w \sin \theta_w = R n_m \sin \theta_m \quad (2.77)$$

One can readily see that diffraction calculations using the mask dimensions and the mask side numerical aperture produce the same results as calculations using the mask scaled to wafer dimensions when using the wafer side NA . Additionally, all longitudinal

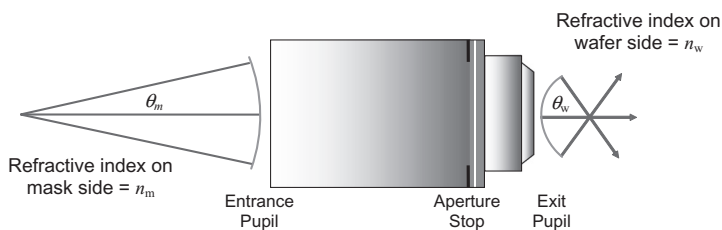


Figure 2.11 An imaging lens with reduction R scales both the lateral dimensions and the sine of the angles by R

(z , the focus direction) dimensions at the wafer are reduced by approximately R^2 compared to focus variations at the mask (see Chapter 3 for a more exact description).

Reduction (or magnification) in an imaging system adds an interesting complication. Light entering the objective lens can be assumed to leave the lens with no loss in energy (the lossless lens assumption). However, if there is reduction or magnification in the lens, the *intensity* distribution of the light entering will be different from that leaving since the intensity is the energy spread over a changing area. The result is a radiometric correction well known in optics.³ Conservation of energy requires that the amplitude of the electric field passing through the pupil at a wafer-side diffraction angle θ_w (corresponding to an angle at the mask side of θ_m) be modified by the radiometric correction

$$\text{Radiometric Correction} = \sqrt{\frac{\cos \theta_m}{\cos \theta_w}} = \left(\frac{1 - \left(\frac{n_w}{n_m} \right)^2 \frac{\sin^2 \theta_w}{R^2}}{1 - \sin^2 \theta_w} \right)^{0.25} \quad (2.78)$$

where R is the reduction factor (for example, 4.0 for a 4 $\times$ reduction imaging tool, see Figure 2.12) and Equation (2.77) was used to convert mask-side angles to the wafer side. This radiometric correction can be incorporated directly into the lens pupil function in terms of the wafer-side spatial frequencies:

$$P(f_x, f_y) = P_{\text{ideal}}(f_x, f_y) \left(\frac{1 - \frac{\lambda^2(f_x^2 + f_y^2)}{n_m^2 R^2}}{1 - \frac{\lambda^2(f_x^2 + f_y^2)}{n_w^2}} \right)^{0.25} \quad (2.79)$$

Note that for R of about four and higher, the radiometric correction can be well approximated as

$$P(f_x, f_y) \approx P_{\text{ideal}}(f_x, f_y) \left(1 - \frac{\lambda^2(f_x^2 + f_y^2)}{n_w^2} \right)^{-0.25} \quad (2.80)$$

The existence of this radiometric correction phenomenon gives rise to an interesting question about image intensity normalization. In essence, for aerial image normalization,

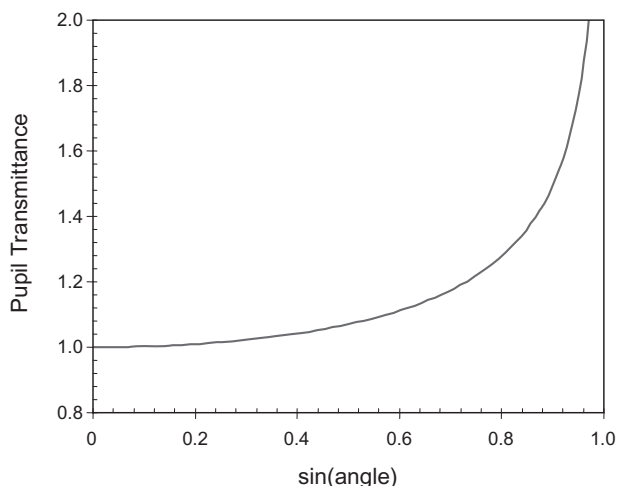


Figure 2.12 A plot of the radiometric correction as a function of the wafer-side diffraction angle ($R = 4$, $n_w/n_m = 1$)

there are two choices: wafer level and mask level normalization. For example, one could define a unit relative intensity as the intensity obtained at the wafer for a large open-field exposure (blank glass mask in the reticle plane). Alternatively, one could define the unit intensity as the intensity transmitted by a large open area on a mask (and suitably multiplied by the reduction ratio squared to take the area change between mask and wafer into account). If the radiometric correction is ignored, then these normalization approaches are identical by invocation of the ‘lossless lens’ approximation: the energy leaving the mask is 100% transmitted to the wafer and thus the intensity just below the mask is identical to the intensity striking the wafer for a blank quartz mask. However, if the radiometric correction is taken into account, this simple, intuitive relationship no longer holds (see Figure 2.13). Conservation of energy requires that, for a system with magnification or reduction, the intensity of light will change from one side of the lens to the other if that light is traveling at a non-normal direction. As a result, when the radiometric correction is taken into account, the calculated aerial images will be different for a system with a nonunit reduction ratio depending on whether mask-side or wafer-side normalization is used. The location of the energy sensor on the imaging tool, whether at the mask plane or the wafer plane, dictates which normalization scheme is most appropriate (most lithographic imaging tools have their dose sensors at the wafer plane).

More importantly than intensity normalization, the radiometric correction impacts the shape and quality of an aerial image. By increasing the pupil transmittance for high spatial frequencies, the radiometric correction can actually improve image quality for some masks and illuminators by making the higher-frequency diffraction orders (which carry the information about the smallest features on the mask) brighter.

Of course, the purpose of reduction in imaging has nothing to do with the radiometric correction. Masks are made larger by some multiple in order to simplify the making of the mask. Making masks with features that are $4\times$ bigger than on the wafer is far easier

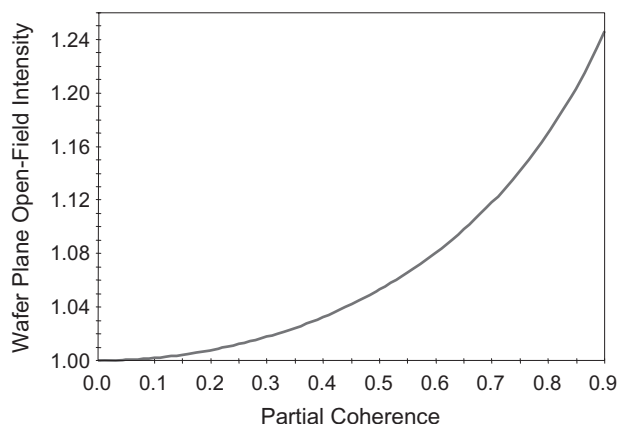


Figure 2.13 One impact of the radiometric correction: the intensity at the wafer plane for a unit amplitude mask illumination as a function of the partial coherence factor ($R = 4$, $NA = 0.9$)

than making $1\times$ features on the mask. However, there are two important trade-offs with increasing reduction ratio. For a given image field size, the physical dimensions of the mask grow as the reduction ratio. For a 25×25 mm image field and a $4\times$ mask, the resulting mask size would need to accommodate a 100×100 mm region. If the reduction ratio were made larger, the mask size could become too large to handle effectively, especially with respect to maintaining proper registration of the features over the larger mask area. Additionally, higher reduction ratios lead to more complex lens designs (the symmetry of $1\times$ imaging leads to the simplest lens designs).

2.3 Partial Coherence

Although we have completely described the behavior of one simple, ideal imaging system, we must add an important complication before we describe the operation of a projection system for lithography. So far, we have assumed that the mask is illuminated by *spatially coherent* light. Coherent illumination means simply that the light striking the mask arrives from only one direction. We have further assumed that the coherent illumination on the mask is normally incident. But real lithographic tools use a more complicated form of illumination, where light strikes the mask from a range of angles.

(As a note, the coherence described here is a *spatial* coherence, as opposed to the related but different concept of *temporal* coherence. Spatial coherence can be thought of as the phase relationship between light at two points in space at one instance in time. Temporal coherence looks at light at one point in space but at different times. While temporal coherence often entails spatial coherence, the existence of spatial coherence usually does not imply anything about the temporal coherence of the light. Unfortunately, most technical literature will refer to one or the other of these properties as simply ‘coherence’. Thus, in this book, coherence means spatial coherence unless specifically described otherwise.)

2.3.1 Oblique Illumination

For spatially coherent illumination (a plane wave of illumination normally incident on the mask), diffraction resulted in a pattern that was centered in the entrance to the objective lens. What would happen if we changed the direction of the illumination so that the plane wave struck the mask at some angle θ' ? While the incident electric field will still have a magnitude of 1, the phase of this plane wave will vary linearly across the mask. For a plane wave tilted relative to the x -axis,

$$E_i(x,y) = e^{i2\pi \sin \theta' x / \lambda} \quad (2.81)$$

Under the Kirchhoff boundary condition assumption, the effect is simply to shift the position of the diffraction pattern with respect to the lens aperture (in terms of spatial frequency, the amount shifted is $f'_x = \sin \theta' / \lambda$), as seen in Figure 2.14 (see Problem 2.16). Recalling that only the portion of the diffraction pattern passing through the lens aperture is used to form the image, it is quite apparent that this shift in the position of the diffraction pattern can have a profound effect on the resulting image. Letting f'_x and f'_y be the shift in the spatial frequency due to the tilted illumination, Equation (2.61) becomes

$$\begin{aligned} E(x,y,f'_x,f'_y) &= \mathcal{F}^{-1}\{T_m(f_x - f'_x, f_y - f'_y)P(f_x, f_y)\} \\ I(x,y,f'_x,f'_y) &= |E(x,y,f'_x,f'_y)|^2 \end{aligned} \quad (2.82)$$

The impact of shifting the diffraction pattern within the objective lens pupil on the quality of the resulting image is hard to generalize. In some cases, the result might be loss of an important diffraction order that otherwise would have made it through the lens. In another case, the shift might enable an otherwise lost diffraction order to be captured. The impact will be dependent on the incident angle, the mask pattern (for example, the pitch) and on the spatial frequency cutoff, NA/λ . Although the lithographic impact is complicated, some general rules will be developed in Chapter 10.

One of the most important lithographic consequences of tilting the illumination is a change in the resolution. Equation (2.57) showed that for normally incident illumination, the smallest half-pitch that could be imaged was $0.5\lambda/NA$. By tilting the illumination so

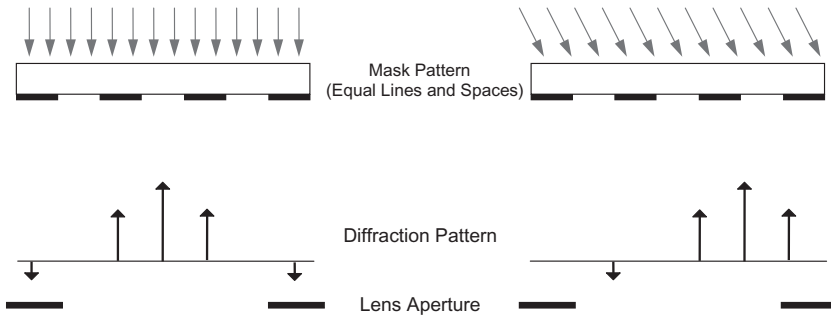


Figure 2.14 The effect of changing the angle of incidence of plane wave illumination on the diffraction pattern is simply to shift its position in the lens aperture. A positive illumination tilt angle is depicted here

that the zero order of light just passes through the right edge of the lens, the pitch can be reduced until the -1 st order just passes through the left edge of the lens. Thus, the distance between diffraction orders ($1/p$) is equal to the diameter of the lens ($2NA/\lambda$) and the smallest half-pitch becomes

$$R = 0.25 \frac{\lambda}{NA} \quad (2.83)$$

Tilting the illumination has the potential for improving resolution by a factor of 2.

2.3.2 Partially Coherent Illumination

If the illumination of the mask is composed of light coming from a range of angles rather than just one angle, the illumination is called *partially coherent*. If one angle of illumination causes a shift in the diffraction pattern, a range of angles will cause a range of shifts, resulting in broadened diffraction orders (Figure 2.15). The most common type of partially coherent source (called a ‘conventional’ source) provides a uniform range of angles illuminating the mask, with each angle having equal intensity. One can characterize the range of angles used for the illumination in several ways, but the most common is the *partial coherence factor*, σ (also called the degree of partial coherence, the pupil filling function, or just the partial coherence). The partial coherence is defined as the sine of the half-angle of the illumination cone divided by the objective lens numerical aperture:

$$\sigma = \frac{n \sin(\theta_{\max})}{NA} = \frac{\text{source diameter}}{\text{lens diameter}} \quad (2.84)$$

(where a wafer-side equivalent illumination angle is used, calculated from the Abbe sine condition). It is thus a measure of the angular range of the illumination relative to the angular acceptance of the lens. As we shall see in section 2.3.7, the illumination system creates an image of the light source at the objective lens pupil. Thus, a conventional illumination source will appear as a circle of uniform intensity when imaged into the pupil

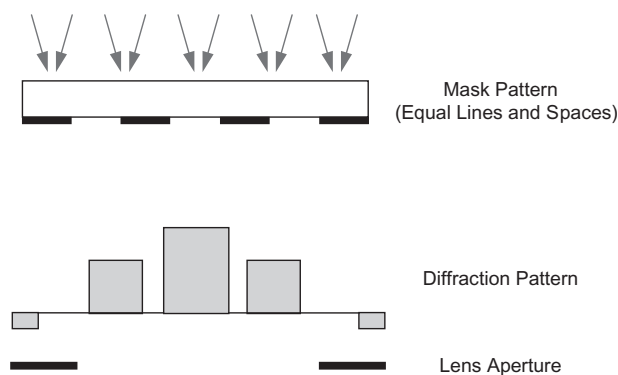


Figure 2.15 The diffraction pattern is broadened by the use of partially coherent illumination (plane waves over a range of angles striking the mask)

(Figure 2.16). Another form of the definition of the partial coherence factor of a conventional source is the diameter of the source disk divided by the diameter of the objective lens aperture. Finally, if the range of angles striking the mask extends from -90° to 90° (that is, all possible angles), the illumination is said to be *incoherent*. Table 2.2 helps define the terminology of spatial coherence. Note that the maximum partial coherence factor (assuming the mask remains in air) is R/NA , where R is the reduction ratio. Thus, perfectly incoherent illumination is possible only in the limit of very small numerical apertures. However, from an imaging perspective any partial coherence factor greater than 2 is essentially incoherent and is usually treated as such.

An arbitrarily shaped illumination source can be described as an *extended source*, where the source shape is divided into a collection of individual point sources. Each point is a source of spatially coherent illumination, producing a plane wave striking the mask at one angle, and resulting in an aerial image given by Equation (2.82). Two point sources from the extended source, however, do not interact coherently with each other. Thus, the contributions of these two sources must be added to each other incoherently (that is, the intensities are added together). Illumination systems for lithographic tools are carefully designed to have this property. If two spatially separated points on the source were phase-related, light from those points would interfere at the mask plane, creating nonuniform illumination (the phenomenon of speckle is an example of such coherent interaction of source points).

The full aerial image is determined by calculating the coherent aerial image from each point on the source, and then integrating the intensity over the source (an approach known as the Abbe method of partially coherent image calculation). The source can be defined

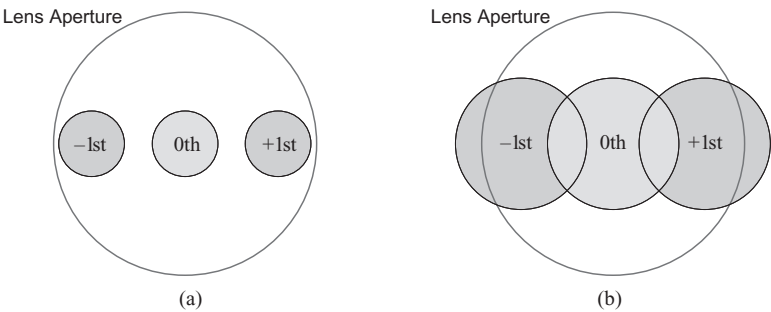


Figure 2.16 Top-down view of the lens aperture showing the diffraction pattern of a line/space pattern when partially coherent illumination is used: (a) $\sigma = 0.25$; and (b) $\sigma = 0.5$

Table 2.2 Partial coherence types (conventional illumination)

Illumination Type	Partial Coherence Factor	Source Shape
Coherent	$\sigma = 0$	Point source
Incoherent	$\sigma = \infty$	Infinite size source
Partially Coherent	$0 < \sigma < \infty$ (but generally $0 < \sigma < 1$)	Circular disk-shaped source

by a source function, $S(f'_x, f'_y)$, which is just the intensity of the source as a function of position (or angle). The total intensity of the image is then

$$I_{\text{total}}(x, y) = \frac{\iint I(x, y, f'_x, f'_y) S(f'_x, f'_y) df'_x df'_y}{\iint S(f'_x, f'_y) df'_x df'_y} \quad (2.85)$$

Often, the pupil coordinates (f_x, f_y) are normalized in the same way as the partial coherence factor, creating pupil coordinates in 'sigma' space. These coordinates are

$$\sigma_x = \frac{f_x \lambda}{NA}, \quad \sigma_y = \frac{f_y \lambda}{NA} \quad (2.86)$$

In sigma coordinates, the lens pupil is always the unit circle. Also, the source shape function S can be conveniently normalized as

$$\tilde{S}(\sigma'_x, \sigma'_y) = \frac{S(\sigma'_x, \sigma'_y)}{\iint S(\sigma'_x, \sigma'_y) d\sigma'_x d\sigma'_y} \quad (2.87)$$

so that

$$I_{\text{total}}(x, y) = \iint I(x, y, \sigma'_x, \sigma'_y) \tilde{S}(\sigma'_x, \sigma'_y) d\sigma'_x d\sigma'_y \quad (2.88)$$

The examples of Figure 2.16 illustrate an important difficulty in calculating images with partially coherent illumination. In Figure 2.16a, all of the zero and first diffraction orders are captured. When Equation (2.85) is evaluated, the aerial image for each source point, $I(x, y, f'_x, f'_y)$, will be identical, being made up of three diffraction orders. Thus, the partially coherent image is equal to the coherent image. For the case shown in Figure 2.16b, however, there is a very different result. For some source points, only one of the first orders is captured and thus a different image results.

This idea can be expressed more clearly using Figure 2.17. Again showing a case of imaging equal lines and spaces where only the zero and the two first diffraction orders are captured, the black region in the figure shows all of the source points where three

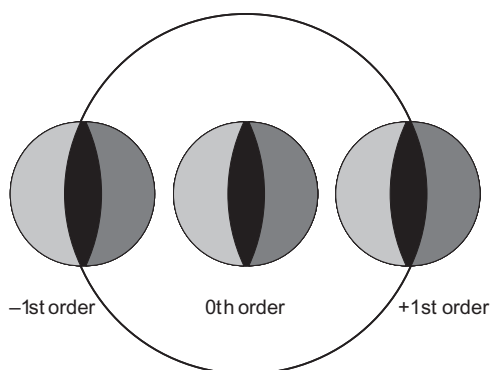


Figure 2.17 Example of dense line/space imaging where only the zero and first diffraction orders are used. Black represents three-beam imaging, lighter and darker grays show the area of two-beam imaging

diffraction orders are captured. This region of the source produces ‘three beam’ imaging, equal to the coherent aerial image of Equation (2.66). The gray areas, however, result in only the zero order and one first order captured by the lens (the light gray area forms an image from the 0 and +1st orders, the dark gray area forms an image from the 0 and –1st orders). These areas form a ‘two-beam’ image, given by

$$I(x) = a_0^2 + a_1^2 + 2a_0a_1 \cos(2\pi x/p) \quad (2.89)$$

The final image will be a weighted average of the two-beam and three-beam images, where the weights are their respective source intensity fractions. Thus, for the case of a uniform intensity source shape and a repeating pattern of lines and spaces, the solution to the partially coherent imaging integrals of Equation (2.85) becomes a geometry problem of calculating the overlapping areas of the various circles for the source and the lens.

This approach, often called the Kintner method,⁴ is readily solvable for the case of a circularly shaped source. Every image takes the form of

$$I(x) = s_1 I_{1\text{-beam}} + s_2 I_{2\text{-beam}} + s_3 I_{3\text{-beam}} \quad (2.90)$$

where

$$\begin{aligned} I_{1\text{-beam}} &= a_0^2 \\ I_{2\text{-beam}} &= a_0^2 + a_1^2 + 2a_0a_1 \cos(2\pi x/p) \\ I_{3\text{-beam}} &= a_0^2 + 4a_0a_1 \cos(2\pi x/p) + 4a_1^2 \cos^2(2\pi x/p) \end{aligned}$$

The weights s_i take on different values for different regimes of pitch and partial coherence, as given in Table 2.3, and always add up to 1. This table makes use of the following values (making sure that the inverse tangent is defined over the range from 0 to π):

$$\chi = \frac{1 + \left(\frac{\lambda}{pNA} \right)^2 - \sigma^2}{\frac{2\lambda}{pNA}} \quad (2.91)$$

Table 2.3 Kintner weights for conventional source imaging of small lines and spaces

Regime	Description	s_1	s_2	s_3
$\frac{\lambda}{pNA} \leq 1 - \sigma$	All three-beam imaging	0	0	1
$1 - \sigma \leq \frac{\lambda}{pNA} \leq \sqrt{1 - \sigma^2}$	Combination of two- and three-beam imaging	0	2γ	$1 - 2\gamma$
$\sqrt{1 - \sigma^2} \leq \frac{\lambda}{pNA} \leq 1$	Combination of one-, two- and three-beam imaging	$\eta + 2\gamma - 1$	$2(1 - \gamma - \eta)$	η
$1 \leq \frac{\lambda}{pNA} \leq 1 + \sigma$	Combination of one- and two-beam imaging	$2\gamma - 1$	$2(1 - \gamma)$	0
$1 + \sigma \leq \frac{\lambda}{pNA}$	Only the zero order is captured, all one-beam	1	0	0

$$\gamma = \frac{\sigma^2 \tan^{-1} \left(\frac{\sqrt{1-\chi^2}}{\chi - \frac{\lambda}{pNA}} \right) - \cos^{-1}(\chi) + \frac{\lambda}{pNA} \sqrt{1-\chi^2}}{\pi \sigma^2} \quad (2.92)$$

$$\eta = \frac{2 \left[\cos^{-1} \left(\frac{\lambda}{pNA} \right) - \frac{\lambda}{pNA} \sqrt{1 - \left(\frac{\lambda}{pNA} \right)^2} \right]}{\pi \sigma^2} \quad (2.93)$$

2.3.3 Hopkins Approach to Partial Coherence

H.H. Hopkins reformulated the Abbe extended source method for partially coherent image calculations in an interesting and occasionally useful way.^{5,6} Equation (2.82) gives the electric field for a single source point by shifting the diffraction pattern relative to a fixed pupil. A change of variables will show that this is exactly equivalent to fixing the diffraction pattern and shifting the pupil by the same amount:

$$E(x, f'_x) = \int_{-\infty}^{\infty} P(f_x + f'_x) T_m(f_x) e^{2\pi i(f_x + f'_x)x} df_x \quad (2.94)$$

where only one dimension is shown for simplicity. Calculating the image intensity for this single source point as the product of the electric field and its complex conjugate,

$$\begin{aligned} I(x, f'_x) &= E(x, f'_x) E^*(x, f'_x) \\ &= \left[\int_{-\infty}^{\infty} P(f_x + f'_x) T_m(f_x) e^{2\pi i(f_x + f'_x)x} df_x \right] \left[\int_{-\infty}^{\infty} P^*(\tilde{f}_x + f'_x) T_m^*(\tilde{f}_x) e^{-2\pi i(\tilde{f}_x + f'_x)x} d\tilde{f}_x \right] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} P(f_x + f'_x) P^*(\tilde{f}_x + f'_x) T_m(f_x) T_m^*(\tilde{f}_x) e^{2\pi i(f_x - \tilde{f}_x)x} df_x d\tilde{f}_x \end{aligned} \quad (2.95)$$

The next step in the Abbe extended source method is to integrate over all of the source points as in Equation (2.85) or (2.88). The critical step in the Hopkins formulation is to change the order of integration and to integrate over the source first, before carrying out the inverse Fourier transform integrals of Equation (2.95). Since only the pupil functions in Equation (2.95) depend on the source point, The integration over the source gives an intermediate variable

$$TCC(f_x, \tilde{f}_x) = \int_{-\infty}^{\infty} P(f_x + f'_x) P^*(\tilde{f}_x + f'_x) \tilde{S}(f'_x) df'_x \quad (2.96)$$

so that

$$I_{\text{total}}(x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} TCC(f_x, \tilde{f}_x) T_m(f_x) T_m^*(\tilde{f}_x) e^{2\pi i(f_x - \tilde{f}_x)x} df_x d\tilde{f}_x \quad (2.97)$$

The intermediate variable TCC in Equation (2.96) is called the *transmission cross-coefficient*. It has the useful property of being independent of the mask transmittance function – it depends only on the pupil function and the source shape. For applications where the pupil and source functions are fixed, but images must be calculated for a wide range for different masks, precalculation of the TCC function can greatly speed up image calculations. Optical proximity correction (OPC), discussed in Chapter 10, is one example where this approach becomes useful. Physically, the TCC can be thought of as an intensity-based description of how two separate pupil points interact when forming an image. Of course, the one-dimensional formulations above can be easily extended to two dimensions.

2.3.4 Sum of Coherent Sources Approach

An approximate solution to the Hopkins imaging equations called *Sum of Coherent Sources* (SOCS) provides extremely fast computation times and is thus very useful for some imaging calculation like those used for optical proximity correction.⁷ To begin, we will convert the final Hopkins equation (2.97) from the spatial frequency domain to the spatial domain. From the definition of the Fourier transform of the mask transmittance,

$$T_m(f_x) = \int_{-\infty}^{\infty} t_m(x) e^{-i2\pi f_x x} dx, T_m * (\tilde{f}_x) = \int_{-\infty}^{\infty} t_m^*(x) e^{i2\pi \tilde{f}_x x} dx \quad (2.98)$$

Also, noting that the electric field point spread function is the Fourier transform of the pupil function,

$$h(x) = \int_{-\infty}^{\infty} P(f_x) e^{-i2\pi f_x x} df_x \quad (2.99)$$

and defining the *mutual coherence function* as the Fourier transform of the normalized source shape,

$$J_0(x) = \int_{-\infty}^{\infty} \tilde{S}(f_x) e^{-i2\pi f_x x} df_x \quad (2.100)$$

the Hopkins imaging integral can be converted to an integration over x (that is, an integration over the image plane) rather than an integration over the pupil plane:

$$I(x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} J_0(x' - x'') h(x - x') h^*(x - x'') t_m(x') t_m^*(x'') dx' dx'' \quad (2.101)$$

Defining a function W that depends only on the source and the pupil,

$$W(a, b) = J_0(a - b) h(a) h^*(b) \quad (2.102)$$

this new Hopkins integral becomes

$$I(x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} W(x - x', x - x'') t_m(x') t_m^*(x'') dx' dx'' \quad (2.103)$$

So far, there is no real advantage to this formulation of the Hopkins imaging equation. Let us now propose an interesting decomposition for W as the weighted sum of its eigenvectors ϕ_n :

$$W(a,b) = \sum_{n=1}^{\infty} \lambda_n \phi_n(a) \phi_n^*(b) \quad (2.104)$$

where λ_n is the eigenvalue for the eigenvector ϕ_n . Expressed in this way, the Hopkins integral becomes

$$\begin{aligned} I(x) &= \sum_{n=1}^{\infty} \lambda_n \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \phi_n(x-x') \phi_n^*(x-x'') t_m(x') t_m^*(x'') dx' dx'' \\ &= \sum_{n=1}^{\infty} \lambda_n \int_{-\infty}^{\infty} \phi_n(x-x') t_m(x') dx' \int_{-\infty}^{\infty} \phi_n^*(x-x'') t_m^*(x'') dx'' \\ &= \sum_{n=1}^{\infty} \lambda_n |\phi_n(x) \otimes t_m(x)|^2 \end{aligned} \quad (2.105)$$

Thus, for a given source shape and pupil function, a set of eigenvectors and eigenvalues can be determined, each eigenvector playing the role of a coherent PSF-like function. These eigenvectors are then convolved with the mask transmittance function to give an electric field at the wafer plane. Converting to intensity, the weighted sum of these image components becomes the total image. The eigen functions also have the convenient property of being orthogonal to each other. Note that for the case of coherent illumination, $J_0(x) = 1$, $\lambda_1 = 1$, $\phi_1 = h(x)$ and $\lambda_n = 0$ for $n > 1$. In fact, one simple decomposition is to let each n represent a coherent point on the source, so that λ_n is the intensity of the source point and ϕ_n is just the coherent PSF with a phase tilt corresponding to the position of that source point. Thought of in this way, Equation (2.105) is just an expression of the Abbe extended source formulation expressed as convolutions in the image plane rather than products in the pupil plane.

Of course, for any real calculation of an image using this approach, the infinite summation must be cut off at some number N :

$$I(x) \approx \sum_{n=1}^N \lambda_n |\phi_n(x) \otimes t_m(x)|^2 \quad (2.106)$$

where higher values of N lead to greater accuracy in the calculation. The key insight into the use of SOCS is to find a set of eigen functions that allows N to be minimized. Letting each term in the summation represent one source point often requires hundreds of terms to produce the desired accuracy. By cleverly choosing a different set of eigen functions, the required number of terms in the summation can be reduced by an order of magnitude or so. However, since a new set of eigen functions and eigenvalues must be determined every time the source or the pupil function is changed, this approach tends to be used only for cases where the source and pupil are fixed and images for a large number of mask patterns must be determined.

2.3.5 Off-Axis Illumination

Off-axis illumination refers to any illumination shape that significantly reduces or eliminates the ‘on-axis’ component of the illumination, that is, the light striking the mask at near normal incidence (Figure 2.14). By tilting the illumination away from normal incidence, the diffraction pattern of the mask is shifted within the objective lens. While all the reasons why this is desirable are not yet obvious (resolution is one benefit, but another involves depth of focus, a concept not yet discussed), there are many lithographic applications where off-axis illumination is preferred. The most popular illumination shapes, besides the conventional disk shape, are annular illumination, some form of quadrupole illumination, and dipole illumination, although many other shapes are used for special applications (see Chapter 10).

The idealized source shapes pictured in Figure 2.18 can be thought of as ‘top-hat’ illuminators: the intensity is one inside the source shape and zero outside of it. However, since these source shapes are created using a variety of optical components, the real source distributions of a real projection tool can be significantly more complicated. Figure 2.19 shows some actual measured source shapes from two different production lithographic

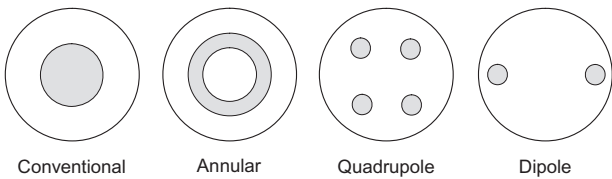


Figure 2.18 Conventional illumination source shape, as well as the most popular off-axis illumination schemes, plotted in spatial frequency coordinates. The outer circle in each diagram shows the cutoff frequency of the lens

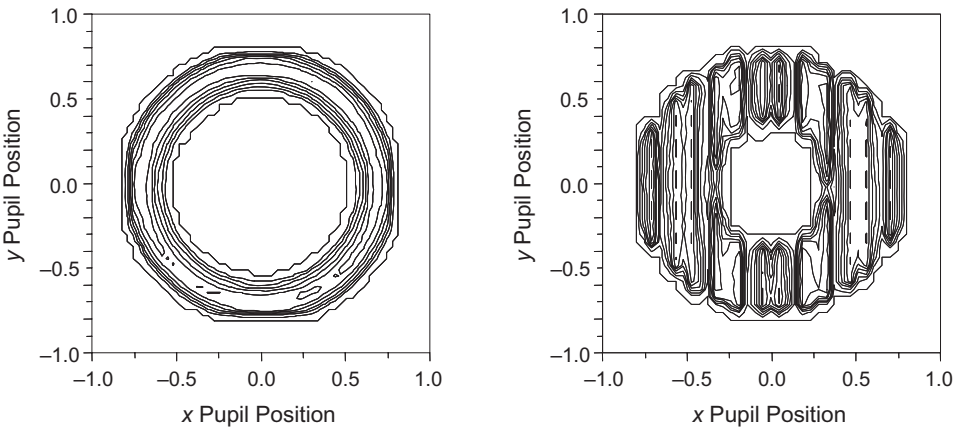


Figure 2.19 Examples of measured annular source shapes (contours of constant intensity) from two different lithographic projection tools showing a more complicated source distribution than the idealized ‘top-hat’ distributions

exposure tools. While both of these source shapes are annular, they exhibit a gradual falloff of intensity with pupil radius rather than a sharp cutoff, and a fine structure that is specific to the optical method of generating the shape.

In many cases, the real source shape can be approximated by the ‘designed’ source shape (the idealized top-hat illumination) convolved with an illumination point spread function, to account for the finite resolution of the illumination system. Often, the illumination point spread function can reasonably be approximated as a Gaussian.

2.3.6 Imaging Example: Dense Array of Lines and Spaces Under Annular Illumination

Off-axis illumination, with annular illumination being a common example, is most often used to push the resolution limits of small, dense patterns. Thus, for the case of a dense array of lines and spaces, these small features will generally only admit the zero and $\pm$ first diffraction orders through the lens (Figure 2.20). As before, the image will depend on the fraction of the first order that is captured by the lens, and whether this fraction results in only two-beam imaging, or a combination of two-beam and three-beam imaging.

While a full Kintner-like solution is certainly possible (see section 2.3.2), here we will develop a simplified approximate solution that will be reasonably accurate whenever the width of the annulus is small. For such a case, the fraction γ of the first order that passes through the lens will be⁸

$$\gamma = \frac{1}{\pi} \cos^{-1} \left\{ \frac{1}{2} \left[\left(\frac{\lambda}{pNA} \right) \frac{1}{\sigma} + \left(\frac{pNA}{\lambda} \right) \left(\sigma - \frac{1}{\sigma} \right) \right] \right\} \quad (2.107)$$

where σ is the radial position of the center of the annulus in σ -coordinates. Using the notation of Equation (2.90), the final image will be the weighted sum of one-beam, two-beam and three-beam images, with weights given in Table 2.4.

2.3.7 Köhler Illumination

In the descriptions of imaging above, the mask was always illuminated by a plane wave, either normally or obliquely incident. Also, in the drawings and analyses of the diffraction patterns given above, the assumption was made that the mask pattern was positioned

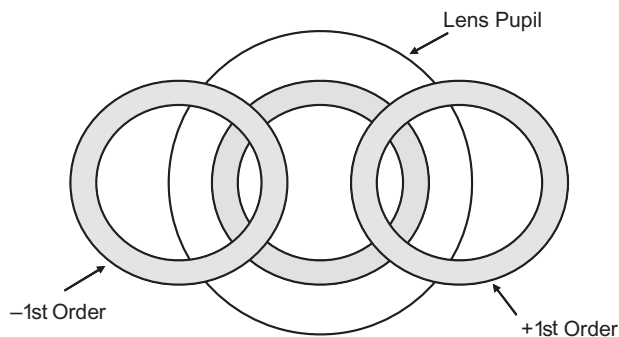


Figure 2.20 Annular illumination example where only the 0th and ± 1 st diffracted orders pass through the imaging lens

Table 2.4 Kintner weights for thin annular source imaging of small lines and spaces

Regime	Description	s_1	s_2	s_3
$\frac{\lambda}{pNA} \leq 1 - \sigma$	All three-beam imaging	0	0	1
$1 - \sigma \leq \frac{\lambda}{pNA} \leq \sqrt{1 - \sigma^2}$	Combination of two- and three-beam imaging	0	$2(1 - \gamma)$	$2\gamma - 1$
$\sqrt{1 - \sigma^2} \leq \frac{\lambda}{pNA} \leq 1 + \sigma$	Combination of one- and two-beam imaging	$1 - 2\gamma$	2γ	0
$1 + \sigma \leq \frac{\lambda}{pNA}$	Only the zero order is captured, all one-beam	1	0	0

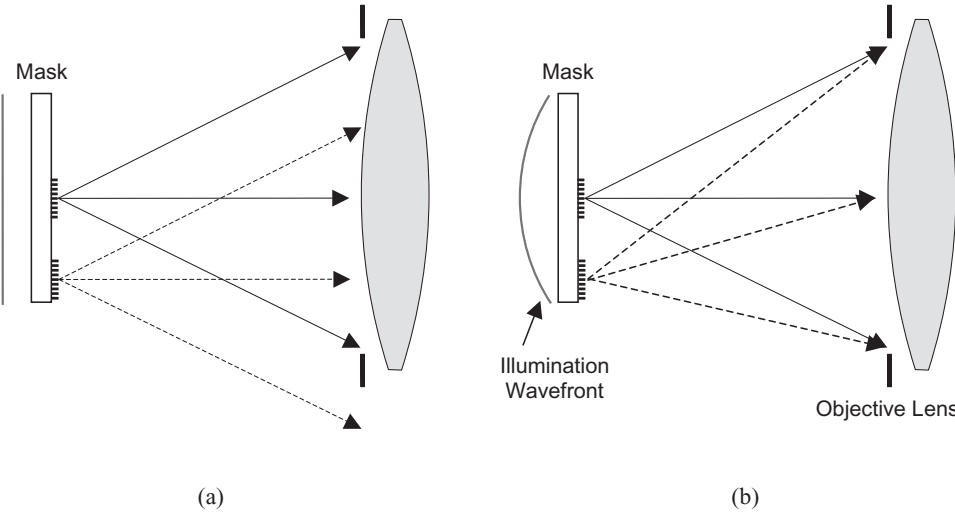


Figure 2.21 The impact of illumination: (a) when illuminated by a plane wave, patterns at the edge of the mask will not produce diffraction patterns centered in the objective lens pupil, but (b) the proper converging spherical wave produces diffraction patterns that are independent of field position

directly on the optical axis (the center line running through the middle of the lens). The mask and the resulting image field, however, are reasonably large so that imaging takes place for mask features that are a long way from the optical axis. This poses a significant problem: diffraction patterns produced by features toward the edge of the mask will produce diffraction patterns that are not centered in the lens (see Figure 2.21a).

The shifting behavior of oblique illumination offers a solution to the problem. It is very desirable that patterns at all positions of the mask be printed with about equal imaging characteristics. Thus, identical mask patterns at different field positions should produce diffraction patterns that are positioned identically within the objective lens pupil. The diffraction pattern for the line/space pattern at the mask edge of Figure 2.21a can be made

to behave identically with the mask pattern on the optical axis by shifting the diffraction pattern up so that it is centered in the lens. This shift, as in section 2.3.1, can be accomplished by tilting the illumination. In fact, at every point on the mask the diffraction patterns can be made to coincide if the illumination is properly tilted for that mask point. The ideal tilt is achieved across the entire mask if the illumination is made to be a spherical wave converging to a point in the entrance pupil of the objective lens (Figure 2.21b).

Note that all of the previous calculations of diffraction patterns assumed a plane wave of illumination striking the mask. If a converging spherical wave is used instead as the illumination, the previous results are still valid so long as the radius of curvature of the spherical wave (that is, the distance from the mask to the objective lens entrance pupil) is sufficiently large compared to the feature dimensions on the mask, so that the spherical wave appears like a plane wave over the region of interest on the mask. This is most certainly true in the Fraunhofer diffraction region. Thus, even though spherical wave illumination will be used, plane wave illumination can still be assumed in the diffraction calculations.

In most lithographic tools today, the various shapes and sizes of illumination are achieved by an optical arrangement known as *Köhler illumination*. Named for August Köhler, developer of illumination systems for Zeiss microscopes in the late 1800s, Köhler illumination uses the condenser lens to form an image of the source at the entrance pupil of the objective lens. Then, the mask is placed precisely at the exit pupil of the condenser lens (Figure 2.22). This arrangement has several very desirable properties. First, the mask is illuminated with converging spherical waves, producing the correct directionality of the incident light over the entire mask as discussed above. Thus, the diffraction pattern produced by a mask pattern will land in the entrance pupil at precisely the same spot, regardless of the position of the mask pattern within the field (i.e. its location on the mask). Second, the uniformity of the illumination intensity over the mask is independent of the uniformity of the intensity of the source itself (since every point on the mask receives light from every point on the source). Thus, Köhler illumination helps ensure uniform imaging for all points on the mask.

An interesting side effect of the use of Köhler illumination is that the focal plane of the source becomes the far field of the diffraction pattern, in the Fraunhofer sense of the far field. That is, the Fresnel diffraction integrals evaluated at the entrance pupil of the objective lens become equal to the Fraunhofer diffraction integrals without further

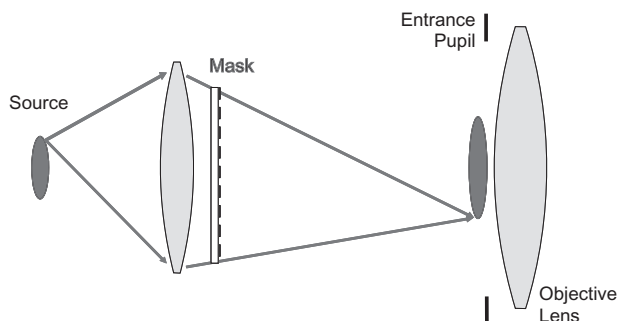


Figure 2.22 Köhler illumination where the source is imaged at the entrance pupil of the objective lens and the mask is placed at the exit pupil of the condenser lens

approximation. By making only the Fresnel diffraction approximations, the diffraction pattern can still be calculated as the Fourier transform of the mask transmittance. In essence, the condenser lens brings the ‘infinite’ diffraction plane in closer, to the image plane of the condenser lens.

2.3.8 Incoherent Illumination

As mentioned above, as the size of the illumination source becomes large relative to the size of the pupil (that is, as σ for an ideal conventional source goes to infinity), the illumination is said to be spatially incoherent. For this case, integration over the source can be greatly simplified. Using the Hopkins formulation, the TCC becomes

$$TCC(f_x, \tilde{f}_x) \propto \int_{-\infty}^{\infty} P(f_x + f'_x) P^*(\tilde{f}_x + f'_x) df'_x = \int_{-\infty}^{\infty} P(f'_x) P^*(\tilde{f}_x - f_x + f'_x) df'_x \quad (2.108)$$

As the final form of this equation indicates, the TCC will only be a function of the difference in spatial frequencies $\tilde{f}_x - f_x$. Thus,

$$TCC(\tilde{f}_x - f_x) \propto P(f_x) \otimes P^*(f_x) \quad (2.109)$$

where the proportionality is made into an equality through proper normalization. For incoherent illumination, the TCC is called the *Optical Transfer Function* (OTF). The magnitude the OTF is called the *Modulation Transfer Function* (MTF). It is usually normalized by dividing by the area of the pupil. For an ideal circular pupil, this convolution can be easily carried out (being interpreted geometrically as the area overlap between shifted circles, Figure 2.23) giving

$$MTF(f) = \begin{cases} \frac{2}{\pi} \left[\cos^{-1} \left(\frac{f}{2NA/\lambda} \right) - \frac{f}{2NA/\lambda} \sqrt{1 - \left(\frac{f}{2NA/\lambda} \right)^2} \right], & f \leq 2 \frac{NA}{\lambda} \\ 0, & \text{otherwise} \end{cases} \quad (2.110)$$

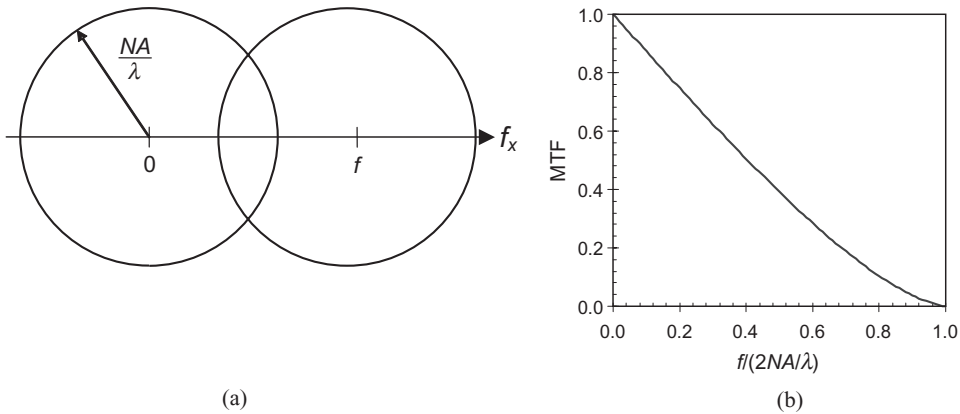


Figure 2.23 The $MTF(f)$ is the overlapping area of two circles of diameter NA/λ separated by an amount f , as illustrated in (a) and graphed in (b)

where $f = \tilde{f}_x - f_x$ for the case of our 1D features. One can see that the cutoff frequency for the incoherent MTF is $2NA/\lambda$, corresponding to the zero order at one edge of the lens and the first order at the opposite edge.

Using the MTF in Equation (2.97),

$$I_{\text{total}}(x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} MTF(f) T_m(f_x) T_m^*(f_x + f) e^{-2\pi i f x} df_x df \quad (2.111)$$

$$I_{\text{total}}(x) = \int_{-\infty}^{\infty} MTF(f) e^{-2\pi i f x} df \int_{-\infty}^{\infty} T_m(f_x) T_m^*(f_x + f) df_x$$

But, the convolution of the diffraction pattern with its complex conjugate can be easily computed using the convolution theorem of the Fourier transform:

$$\int_{-\infty}^{\infty} T_m(f_x) T_m^*(f_x + f) df_x = \mathcal{F}\{t_m(x) t_m^*(x)\} \quad (2.112)$$

The product of the electric field transmittance of the mask with its complex conjugate is just the intensity transmittance of the mask. Thus, Equation (2.111) becomes

$$I_{\text{total}}(x) = \mathcal{F}^{-1}\{MTF(f_x) IT_m(f_x)\} \quad (2.113)$$

where $IT_m(f_x)$ is the Fourier transform of the intensity transmittance of the mask.

As we saw at the beginning of this chapter, for coherent illumination the electric field image is a linear function of the electric field mask transmittance in Fourier space, with the pupil function as the linear transfer function. Equation (2.113) shows that for incoherent illumination, the intensity image is a linear function of the intensity mask transmittance in Fourier space, with the MTF as the linear transfer function. For partially coherent illumination, the relationship between the image and the object is nonlinear.

As an example of an incoherent image, consider a binary pattern of lines and spaces. The resulting aerial image is

$$I(x) = a_0 + 2 \sum_{j=1}^N MTF(j/p) a_j \cos(2\pi j x/p) \quad (2.114)$$

For equal lines and spaces where only the 0 and ± 1 st diffraction orders are captured, the aerial image becomes

$$I(x) = \frac{1}{2} + \frac{2}{\pi} MTF(1/p) \cos(2\pi x/p) \quad (2.115)$$

2.4 Some Imaging Examples

While most of the imaging examples described above were one-dimensional for the sake of mathematical simplicity, the approach described in this chapter can of course be used to calculate more general two-dimensional images. Figure 2.24 shows a few simple examples of two-dimensional aerial images calculated with the lithography simulator PROLITH. Figure 2.25 shows images for an isolated binary edge mask feature as a function of the partial coherence factor.

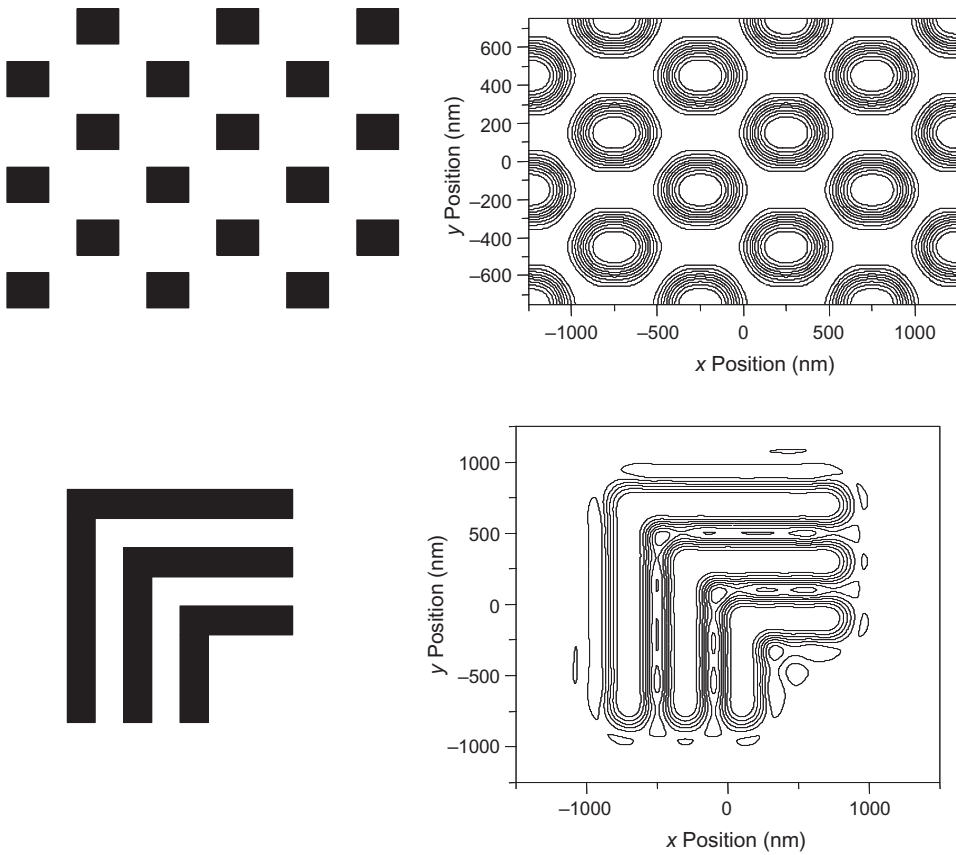


Figure 2.24 Some examples of two-dimensional aerial image calculations (shown as contours of constant intensity), with the mask pattern on the left (dark representing chrome)

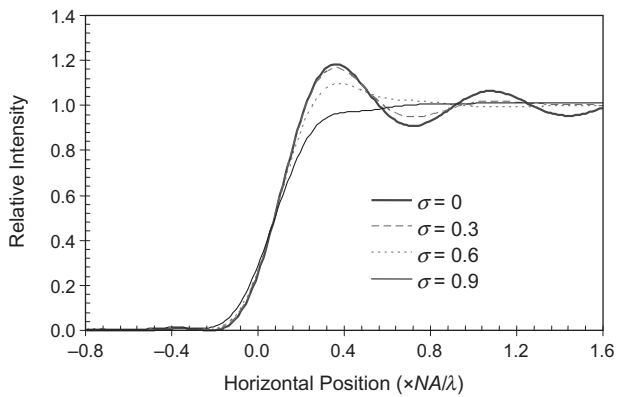


Figure 2.25 Aerial images of an isolated edge as a function of the partial coherence factor

Problems

- 2.1. Derive Equations (2.5).
- 2.2. Show that the spherical wave of Equation (2.27) is in fact a solution to the Helmholtz equation.
- 2.3. Using Equation (2.38), determine the SI units for intensity.
- 2.4. Derive Equations (2.45) for the isolated space and the dense line/space pattern. Some of the theorems from Appendix C may prove useful.
- 2.5. Show that the diffraction pattern for an dense line/space pattern becomes the diffraction pattern for an isolated space as the pitch becomes infinite (for a constant spacewidth w). This is equivalent to proving that

$$\lim_{p \rightarrow \infty} \frac{1}{p} \sum_{j=-\infty}^{\infty} \delta\left(f_x - \frac{j}{p}\right) = 1$$

- 2.6. Complimentary mask features (for example, an isolated line and an isolated space of the same width) are defined by

$$t_m^c(x, y) = 1 - t_m(x, y)$$

Prove that the diffraction patterns of complimentary mask features are given by

$$t_m^c(f_x, f_y) = \delta(f_x, f_y) - T_m(f_x, f_y)$$

Use this expression to derive the diffraction pattern of an isolated line.

- 2.7. Show that the Fourier transform is a linear operation, that is, show that for two functions $f(x, y)$ and $g(x, y)$, and two constants a and b ,

$$\mathcal{F}\{af(x, y) + bg(x, y)\} = a\mathcal{F}\{f(x, y)\} + b\mathcal{F}\{g(x, y)\}$$

- 2.8. Prove the shift theorem of the Fourier transform:

$$\text{If } \mathcal{F}\{g(x, y)\} = G(f_x, f_y), \quad \mathcal{F}\{g(x - a, y - b)\} = G(f_x, f_y) e^{-i2\pi(f_x a + f_y b)}$$

- 2.9. Prove the similarity theorem of the Fourier transform:

$$\text{If } \mathcal{F}\{g(x, y)\} = G(f_x, f_y), \quad \mathcal{F}\{g(ax, by)\} = \frac{1}{|ab|} G\left(\frac{f_x}{a}, \frac{f_y}{b}\right)$$

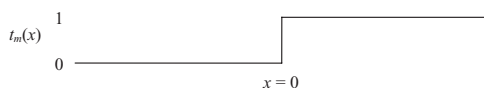
- 2.10. Prove the convolution theorem of the Fourier transform:

$$\text{If } \mathcal{F}\{g(x, y)\} = G(f_x, f_y) \quad \text{and} \quad \mathcal{F}\{h(x, y)\} = H(f_x, f_y),$$

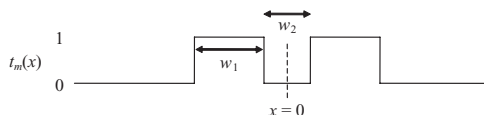
$$\mathcal{F}\left\{\int \int_{-\infty}^{\infty} g(\xi, \eta) h(x - \xi, y - \eta) d\xi d\eta\right\} = G(f_x, f_y) H(f_x, f_y)$$

- 2.11. Prove the scaling property of the Dirac delta function [in Appendix C, Equation (C.10)].

2.12. Derive the diffraction pattern of an isolated edge:

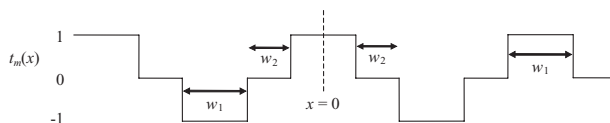


2.13. Derive the diffraction pattern of an isolated double space pattern:



(Hint: Superposition plus the shift theorem may prove useful.)

2.14. Derive the diffraction pattern of an alternating phase shift mask (a pattern of lines and spaces where every other space is shifted in phase by 180° , resulting in a transmittance of -1):



Graph the resulting diffraction pattern out to the ± 5 th diffraction orders for the case of $w_1 = w_2$.

- 2.15. Prove Equation (2.59).
- 2.16. In section 2.3, it was claimed that illuminating a mask with a plane wave at an angle θ' resulted, under the Kirchhoff boundary condition assumption, in a diffraction pattern equal to the normally incident diffraction pattern shifted by an amount $\sin\theta'/\lambda$. Prove this statement using the definition of the Fraunhofer diffraction pattern, Equation (2.44).
- 2.17. Consider a binary pattern of lines and spaces. Which diffraction orders pass through the lens under these circumstances:
- $\lambda = 248 \text{ nm}$, $NA = 0.8$, pitch = 300 nm, on-axis coherent illumination
 - Same as (a), but pitch = 400 nm
 - Same as (b), but illumination tilted by an angle $\sin\theta = 0.5$
- 2.18. Equations (2.65) and (2.67) give the electric field and intensity images for three-beam images, where the 0 and ± 1 st diffraction orders are used to form the image. Derive similar expressions for two-beam images, where only the 0 and +1st diffraction orders are captured by the lens.
- 2.19. For a repeating line/space pattern and coherent illumination, derive expressions for the aerial image intensity at the center of the line and the center of the space as a function of the number of diffraction orders captured.
- 2.20. For $\lambda = 248 \text{ nm}$, $NA = 0.8$ and pitch = 400 nm, below what value of σ is the image entirely made up of three-beam interference? At what σ value does one-beam imaging first appear?

- 2.21. For coherent illumination and a lens of a given NA and λ , derive a formula for the electric field image of an isolated edge. The result will be in terms of the sine integral, defined as

$$\text{Si}(z) = \int_0^z \frac{\sin y}{y} dy$$

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